When Does a Small Model Know to Hand Off? Reading Competence from Full-Duplex Speech Models for Real-Time Escalation

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Abstract

Full-duplex speech models answer in real time with a small on-device backbone, yet many queries exceed its competence. Routing to a large cloud model sounds solved — until one asks *when* the decision must be made: before the first output token, or the wrong answer is already on the air. We ask whether the model's own hidden state carries that warning early enough, and find that the obvious readout is exactly what duplex fine-tuning corrupts: final-layer probes collapse under distribution shift on text input (leave-one-pool-out math AUC 0.93→0.37), consistent with late-layer specialization for duplex turn control, while verbalized self-assessment collapses on audio input. The signal is not gone — it sits one layer band earlier. Mid-network, the same linear probe transfers at 0.93, reads speech input from text-only calibration at 0.86, and an end-of-turn read in the streaming loop reaches AUC 0.887 in $\sim$30 ms — half a second before the first audible word. Run live end-to-end in a turn-based loop, the gated system climbs from 42.2% to 63.3% heard accuracy at 57% escalation, the remaining gap dominated by the transcription uplink rather than the gate; frozen, the same gate improves five of six public speech benchmarks, lifting Speech TriviaQA to 86.0% live against the model's official offline 75.5%. Reading one band earlier, before speech commits, is enough to escalate live.

1 Introduction

A full-duplex speech model listens and speaks in the same forward pass, under a conversational deadline of a few hundred milliseconds [Stivers et al., 2009]. That deadline forces a small on-device backbone — and a small backbone routinely faces queries beyond its competence: long-tail facts, expert knowledge, multi-step reasoning. The natural architecture is selective escalation to a large cloud model, but the duplex setting imposes two constraints that existing routing work does not meet: the decision must come *before the first output token* (once the model starts speaking, a wrong answer is already on the air), and the session — audio context, voice, cache — must stay with the local model, which will speak the expert's answer itself. The question is not whether one can build a router; it is whether the small model knows it will fail *early enough to do anything about it*.

To answer it, we study MiniCPM-o 4.5 [Cui et al., 2026], a 9B omni/duplex fine-tune of Qwen3-8B, against matched-pair controls spanning raw backbones, streaming-omni, audio-only, and frozen-adapter fine-tunes, and evaluate every standard zero-shot competence readout — decode-time entropy, verbalized self-assessment ($p(\mathrm{True})$), and linear probes on hidden states — with judged fail-

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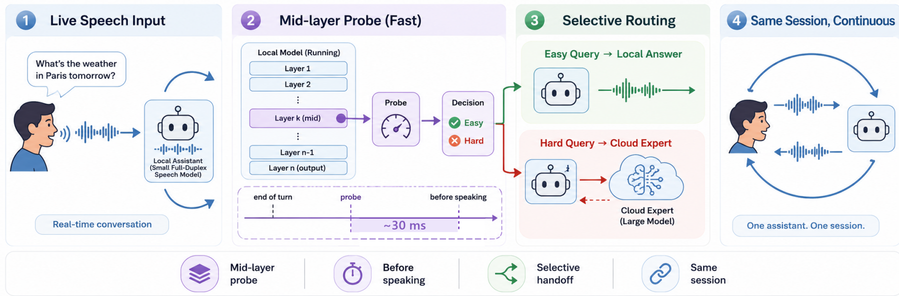

Figure 1: **(1)** Speech streams live into a small duplex-trained local model. **(2)** A mid-network (L22) linear read decides in ∼30 ms — before the first output token — so the expert call is already in flight while local decoding would just be starting. **(3)** Easy queries stay local; hard ones escalate to a cloud expert. **(4)** The *same talker speaks* the expert's answer in one continuous session, improving live spoken sessions (42.2%→63.3%) and, frozen, public speech benchmarks (Speech TriviaQA 66.4%→86.0%).

ure labels on a frozen pool of public-benchmark queries, in matched text and speech conditions. Throughout, *duplex* names the training regime, not the interaction protocol: we study duplex-trained weights driven through a *turn-based* loop whose only decision point is an end-of-turn read (§6); talker-on replay and concurrent listen/speak validations: Appendix H.

Everything runs on a frozen checkpoint — no fine-tuning; the only fitted component is a logistic regression on stored hidden states, calibrated once on text. That minimalism is the point: competing signals read the wrong place (late layers), arrive too late (post-answer verification), or must be trained in (think tokens); the mid-layer read survives all three. Three findings — find the signal, read it in time, show it matters:

1. **Representation: duplex fine-tuning makes the late-layer readouts brittle; a mid-network signal survives.** The conventional final-layer probe looks strong in distribution (AUC 0.82) but is mostly a query-type shortcut: under leave-one-pool-out transfer it *inverts* on held-out math (0.93→0.37). The damage is specific to the late-layer last-token position on text input — absent in raw backbones and non-duplex fine-tunes — consistent with late-layer specialization for duplex turn control. Mid-network (L22 of 36) the same linear read transfers at 0.93, and the mid-network optimum generalizes to a second, architecturally disjoint duplex family (hybrid-Mamba NemotronLabs-VoiceChat-11B).

2. **Modality: the mid-layer signal transfers text → speech in end-to-end models.** A probe calibrated on cheap text data reads speech-input hidden states at 0.86; the control that isolates the mechanism is Freeze-Omni, an adapter on frozen identical weights, where cross-modal transfer is dead (≤0.60). Verbalized self-assessment shows the mirror-image failure — it collapses on audio input — so the mid-layer probe is the one readout that survives both modalities. The transfer holds on real human recordings, not just TTS.

3. **System: the signal is available before the first token and works live, in and out of distribution.** The L22 read costs 20 ms (text) / 45 ms (audio) against time-to-first-token of 36/68 ms; in the streaming loop an end-of-turn read reaches AUC 0.887 in ∼30 ms, half a second before first audible speech. Run live end-to-end — expert consulted mid-conversation, answer injected, talker speaks it — heard accuracy climbs 42.2%→63.3% at 57% escalation, and the frozen gate transfers to six public speech benchmarks with label-free thresholds, lifting Speech TriviaQA to 86.0% against the model's official offline 75.5%.

Extended analyses and controls are in the appendix.

2 Related Work

Full-duplex real-time dialogue. End-to-end spoken dialogue spans native duplex designs — dGSLM [Nguyen et al., 2023], Moshi [Défossez et al., 2024], GLM-4-Voice [Zeng et al., 2024], single-LLM native full-duplex [Fang et al., 2026] — streaming omni models [Xu et al., 2025a, Cui et al., 2026], frozen-backbone adapters [Wang et al., 2024b], and audio fine-tunes [Chu et al., 2024]. Inside these models the standard control interface for turn-taking is a policy *trained into the generator*: a neural-FSM scheme has the LLM emit explicit listen/speak transition tokens [Wang et al., 2024a]; duplex time-slicing learns an idle token that keeps the model silent [Zhang et al., 2024]; SyncLLM aligns the two speech streams with synchronization tokens [Veluri et al., 2024]; LSLM halts generation on a detected-interruption token [Ma et al., 2024]; VITA and SALMONN-omni emit state tokens around the speak/listen decision [Fu et al., 2024, Yu et al., 2024]; DuplexSLA adds a third structured-action channel so tool calls are emitted without pausing speech [Zhang et al., 2026b]; BayLing-Duplex converts a turn-based SpeechLM into a native duplex one with a handful of state tokens and interleaved channels [Fang et al., 2026]; and the checkpoint we probe is itself of this design — MiniCPM-o 4.5 predicts a binary listen/speak token before it generates content [Cui et al., 2026]. A neighboring line moves the same decision into small classifier heads — FlexDuo [Liao et al., 2025], MinMo [Chen et al., 2025], Freeze-Omni's state heads — at 120–250 ms cadences that show a live loop tolerates far more decision latency than our $\sim$30 ms read [cf. Riera et al., 2026, Wu et al., 2026]; a further line tunes these behaviours by RL over interaction-level rewards [Ohashi et al., 2026]. All of these decide *when the local model should speak*; our gate decides the orthogonal question — *whether the local model should be the one answering* — and their shared choice of a trained last-layer readout is exactly the intervention our results show duplex fine-tuning damages.

Does the model know what it knows? $p(\text{True})$ self-evaluation [Kadavath et al., 2022], verbalized uncertainty [Lin et al., 2022], and truthfulness probes on hidden states [Azaria and Mitchell, 2023, Burns et al., 2023, Marks and Tegmark, 2024] established that correctness is readable internally; intermediate layers carry more of it than the output distribution [Orgad et al., 2025]. Semantic entropy [Kuhn et al., 2023, Farquhar et al., 2024] needs multiple sampled generations; Semantic Entropy Probes [Kossen et al., 2024] amortize it into a hidden-state probe — the closest signal to ours, but read post-hoc on text-only half-duplex models. Multimodal uncertainty fusion [Zhang et al., 2026a] asks a downstream question; we ask *where* in a duplex model a modality-robust read survives at all.

Selective escalation and routing. The classic routers are creatures of the turn-based, text-only setting: FrugalGPT [Chen et al., 2023], HybridLLM [Ding et al., 2024], RouteLLM [Ong et al., 2024], and confidence-based cascades [Xue et al., 2025, Hu et al., 2024, Li et al., 2026] make one decision per complete typed query, from query-side features or preference data, before any model has answered; AutoMix [Aggarwal et al., 2024] escalates by *post-answer* self-verification, which a voice deadline forbids. Both assumptions — query-side features, and the large model answering the user directly — fail in full-duplex voice, where the session lives in the small model (query-side ceiling quantified in §4.1). A second family trains the routing decision *into the weights* as special tokens emitted during decoding: Toolformer [Schick et al., 2023] teaches the model to insert API-call tokens by self-supervised fine-tuning; Self-RAG [Asai et al., 2024] fine-tunes the full model on distilled critic labels so reflection tokens trigger retrieval and self-critique; Co-LLM [Shen et al., 2024] and CITER [Zheng et al., 2025] learn a per-token deferral decision to a larger model; pause and thought tokens [Goyal et al., 2024, Zelikman et al., 2024], fused think/no-think modes [Yang et al., 2025, DeepSeek-AI et al., 2025], and the built-in think gate MiniCPM-o 4.5 itself ships [Cui et al., 2026] route *compute* the same way. The design is uniform — extend the vocabulary, then fine-tune the generator until the policy lives in the checkpoint — and so is the cost: supervision must be constructed and the talker fine-tuned, the policy is frozen at training time (a new expert, budget, or threshold means retraining), and in a duplex model fine-tuning the talker is exactly the intervention §4.2 shows damages the last-layer readout. Our gate is the pluggable complement: a logistic read over a frozen checkpoint, refit from stored activations in minutes, its budget moved by a deployment-time threshold, transferring across modality and model family (§7) — and it beats the model's own trained think token at matched budget (Table 9). Learning-to-defer [Madras et al., 2018] and early exit [Schuster et al., 2022] decide inside one model.

Escalating out of a live duplex session. Two concurrent systems do escalate from a small duplex model to a stronger resource, and both differ from us in *what fires the handoff*. MoshiRAG [Chien et al., 2026] pairs a compact duplex interface with selective retrieval, exploiting the lag between speech onset and the first content word to fetch asynchronously and inject the result as embeddings — the trigger is read off the model's own emerging *output*, and what arrives is knowledge for the small model rather than an answer from a larger one. DuplexOmni [Huang et al., 2026] separates a real-time interaction layer from a pluggable thinking layer and trains a think-control token that hands work to it; the policy is once more fine-tuned into the talker. Our gate decides *before* the talker commits to any output, from a frozen checkpoint's internal state, with no talker fine-tuning and a budget moved by a deployment-time threshold.

The gap. The prior work factors cleanly along two axes — {text-based, full-duplex} $\times$ {policy trained in as special tokens, probe read out of frozen internals}. Three cells are dense: text models both train routing tokens and read probes; duplex models train their control policies in as tokens (or heads), including the two systems above that escalate out of a live session. The fourth cell is empty: no prior work we know of reads a *frozen* full-duplex model's internal state for a pre-answer, modality-robust competence signal — and it is empty at exactly the point our results say token-training is least safe, since fine-tuning the talker is what damages the readout those trained policies sit on. That empty cell is this paper: a probe early enough to drive a live small-to-large handoff while preserving the session.

3 Experimental Setup

Target model and matched-pair controls. The system target is **MiniCPM-o 4.5** (9B; omni + full-duplex fine-tune of Qwen3-8B), bf16, greedy decoding, seed 42. To attribute effects to the *kind* of fine-tune rather than the backbone, we assemble ten models spanning raw backbones, a second duplex generation (MiniCPM-o 2.6), a streaming-omni fine-tune (Qwen2.5-Omni-7B), an audio-only fine-tune (Qwen2-Audio-7B), and a frozen-backbone adapter model (Freeze-Omni, LLM weights byte-identical to its raw control); the full roster is Table 2 (appendix). The comparison statistic is always the within-pair difference.

Query pool, labels, and audio arm. 600 queries in five pools, frozen before any gate experiment (360 calibration / 240 test): *trap* = SimpleQA [Wei et al., 2024] (the target fails 100%), *hard-knowledge* = MMLU-Pro [Wang et al., 2024c], *easy-fact* = TriviaQA [Joshi et al., 2017], *easy-chat* = Dolly/Alpaca [Conover et al., 2023, Taori et al., 2023], *hard-math* = GSM8K tail + MATH-500 [Cobbe et al., 2021, Hendrycks et al., 2021, Lightman et al., 2024]; public benchmarks only, no LLM-generated queries. Adequacy labels come from an LLM judge [Zheng et al., 2023] with structured outputs, blind to the gate and validated by a stronger re-judge (agreement 96.2% text / 94.5% audio; every headline number moves $\leq$1 point under the stronger labels); the escalation expert is `gpt-5.5` at low reasoning effort. The audio arm renders the same frozen pool to speech (single-voice TTS, content unchanged — only modality varies), following Spoken-SQuAD / VoiceBench practice [Li et al., 2018, Chen et al., 2024]; audio-side findings are additionally validated on real human recordings (SD-QA, Faisal et al., 2021) and on the external benchmarks' official audio (§7). Judge prompts, pins, and spend: Appendix M–N.

Gate signals (no model training). (i) decode scalars: max token entropy / margin over the first k decode steps; (ii) linear probes: logistic regression on a single hidden state, position $\times$ layer swept, fit on the calibration split only; (iii) verbalized self-evaluation [Kadavath et al., 2022]: $p(\text{True})$-pre asks "would you answer this correctly?" before answering, $p(\text{True})$-post asks about a drafted answer; $P(\text{Yes})$ is read from the first token's distribution.

4 Reading Competence from Duplex Models

4.1 In distribution every readout looks fine; transfer reveals the shortcut

Table 3 compares the candidate signals on MiniCPM-o 4.5. In distribution the final-layer probe looks healthy at AUC 0.822 — but that number is mostly a *query-type* shortcut. The same hidden state all but names which pool a query came from, a pool-identity oracle alone recovers most of the probe's in-distribution margin, and appending true pool labels to the probe's features changes nothing, because the label is already in the representation. The decisive test is therefore leave-one-pool-out (LOPO) transfer, which we treat as the primary metric throughout. Under LOPO the readout does not merely weaken: on held-out math it *inverts*, to AUC 0.372, and it scores a trap pool the model fails 100% of the time at 0.232 — ranking the queries it is certain to get wrong as the safest to keep local. An external query-surface router fails the same way from the other side — never clearing the pool oracle in distribution, collapsing under LOPO, barely moving with two orders of magnitude more data: instance-level competence lives in the model's state, not on the query surface (Appendices B, E).

Verbalized self-assessment transfers differently. Asked *before* answering, $p(\mathrm{True})$ catches exactly the trap pool the probe misses, scoring it 0.945. But timing is everything: asked *after* drafting, the model endorses its own confident-wrong answer and the signal inverts. Post-draft $p(\mathrm{True})$ is the strongest single readout in distribution, at 0.899, and also the one a voice deadline cannot afford — it bills a full draft before the gate can fire.

4.2 The damage is duplex-specific; the signal survives mid-network

Matched pairs attribute the probe failure to the *kind* of fine-tune, not the backbone: the raw Qwen backbone, label-matched to the duplex model's fail rates, holds LOPO math at 0.825 where the duplex model sits at 0.372; streaming-omni and audio-only fine-tunes are undamaged; a second duplex generation (MiniCPM-o 2.6) reproduces the collapse (Appendix B). A layer $\times$ position sweep localizes it (Figure 2): the collapse is specific to (late layer $\times$ last token) on text input — mean-pooled reads survive at the same depth but 0.13–0.15 below the mid-layer peak (LOPO math $\approx$0.80 here, 0.674 on MiniCPM-o 2.6), and mid-network the last-token read recovers to **0.931 at L22**, near the raw backbone.

What breaks at the output. The natural reading — that duplex training re-purposes the last position for turn-control state — is half right at best, and we tested it rather than asserting it (Appendix D). Four candidate mechanisms fail: the cliff does not move when the model is prefilled in speak mode, the inverting direction is orthogonal to the model's own audio-vs-text axis, source pools stay equally decodable at every depth in both the duplex model and its backbone, and once features are standardized the duplex model's representational drift from that backbone does not localize late at all. What does separate them is how the probe's read is *carried* (Figure 6): through mid-network the transferable signal is distributed across directions, and over the final layers of both duplex models — but neither raw backbone — that distributed component decays and inverts, on MiniCPM-o 4.5 while the read concentrates into a handful of dominant directions. Duplex tuning does not delete the competence signal; it stops delivering it to the output position the standard readout uses. Direct tests of the turn-control reading sharpen this without overturning it: the duplex output layer alone stays biased toward emitting control tokens, yet the inverting direction aligns with neither the control vocabulary nor an explicit listen/speak axis, so which duplex objective causes the loss remains open. The mid-network optimum is not a MiniCPM quirk: on NemotronLabs-VoiceChat-11B — a hybrid-Mamba duplex model, architecturally disjoint — the same recipe peaks mid-band at L30–34 of 56 and transfers to the external pools at AUC 0.70–0.79 with a quarter of the calibration data (Appendix C).

Figure 2: The core representational result: LOPO math AUC by layer and position. On the duplex model the standard (final-layer, last-token) read inverts on text input while mid-network layers hold $\geq$0.9; raw backbones show no cliff. The deployed gate reads L22.

4.3 Text-calibrated, speech-deployed — in end-to-end models only

Probes trained on text hidden states and scored on speech-input hidden states locate where the representation becomes modality-shared: early layers are modality-specific (0.54–0.60), and from $\sim$50% depth text$\to$audio transfer plateaus at 0.82–0.87 (0.855 at the deployed L22). This replicates on MiniCPM-o 2.6 and cross-family on Qwen2.5-Omni. The decisive control is **Freeze-Omni**: identical frozen LLM weights with audio attached by an adapter — and cross-modal transfer is dead (0.52–0.60). The modality-shared mid-layer core is *created by training the backbone on the modality*; the cheap recipe "calibrate on text, deploy on speech" is valid for end-to-end models and invalid for adapter-attached ones. Verbalized introspection shows the mirror-image failure: on audio input, pre-answer $p(\mathrm{True})$ collapses on both MiniCPM duplex generations while both non-duplex controls stay intact; controlled arms show the verbal self-check runs over text-token pathways that audio embeddings do not feed — duplicating the question as text restores it — while a linear probe reads the same information off the same forward pass at 0.93 (Appendix B). All audio-side findings replicate on 200 real human recordings (SD-QA), ruling out TTS artifacts. **The mid-layer probe is the one readout that survives every training regime that can converse**; the live system deploys the same L22 read plus two zero-latency pooled positions (the *deployed* probe; Appendix B).

5 From Signal to Gate

The decision exists before decoding starts. The gate is one 4096-d dot product on a hidden state the forward pass already computes. Measured on H100 (wall-clock, prefill start to L22 score, CUDA-synced; network excluded — the cloud call fires asynchronously; protocol: Appendix F), the decision lands in **20 ms (text) / 45 ms (audio)** at P50 (P95 25/104) against same-bench time-to-first-token of 36/68 ms: $T_{\text{gate}} < T_{\text{first token}}$, far inside the 200–300 ms turn-taking budget [Stivers et al., 2009], and the cloud call can launch while the remaining $\sim$40% of prefill and the whole decode still run. Deciding pre-token is optimal, not merely feasible: one could wait out a few decoded tokens and cancel, but decode-time readouts measure *weaker* (max entropy@4 0.696; early-decode hiddens 0.776 vs. 0.828 at the prompt), and every waited millisecond delays the stall onset and the expert launch (Appendix F).

Accuracy–cost tradeoff. On the frozen test split (n=240): small-only 58.8%, big-only (`gpt-5.5`) 91.7%. Gated escalation beats random escalation at every operating point (Figure 7, appendix); the balanced tier at a 33% budget reaches **77.9–78.7%**, recovering $\sim$58% of the small$\to$big gap at one third of the big-only cost. In distribution the gate is statistically on par with a pool-oracle (type-only) router — as the type-shortcut audit predicts — so its case over type routing rests on transfer, where the type shortcut collapses (§4.1) and the mid-layer signal does not. The gate is fit to *local failure*, not expert benefit: a benefit label would bake the current expert's blind spots into a router meant to survive expert swaps, and against the harder label the frozen scores stay usable out of distribution

(Appendix I). Against the model's own built-in reasoning mode, gated-cloud escalation dominates on both accuracy and latency at every rate (Appendix F).

6 Live End-of-Turn Escalation

The streaming loop. The protocol is turn-based: audio streams in 1 s chunks, but the talker speaks only after the user's turn ends and the model's native listen/speak head is not engaged. Per-chunk probe scores are strictly weaker than the offline read (best chunk statistic AUC 0.764) and miss short utterances whose critical entity lands in the final chunk. The fix is an **end-of-turn read**: after the last audio chunk, prefill an assistant turn containing a single space and read L22 at the assistant-start position. This read reaches **AUC 0.887** — above the offline reference (0.843) — in $\sim$30 ms, still before the first output token, and is the loop's only decision point; escalation-rate thresholds are quantiles of this single score. Speculative mid-stream prefetch was measured and dropped (an expert answer to a partial transcript answers the full question only 7–51% of the time; Appendix G). On escalation the talker speaks a stall phrase, its *own* transcription of the audio goes to the expert (`gpt-5.5`, reasoning effort fixed low — higher effort buys zero accuracy on the escalated population at double the latency), and the returned answer is injected as context for the talker to relay. On the deployed channel the relay is clean: model wrong, correct answer injected, comply $= 1.00$ (n=24); more broadly the competence signal also predicts relay compliance (comply 42%$\rightarrow$75% across gate-score quartiles) — a model that knows it doesn't know is willing to listen. Conflicting-injection and framing controls: Appendix G.

The live result. We ran 240 frozen test queries $\times$ four escalation tiers, each a complete turn-based spoken session, scored on *heard accuracy* — what a user would actually have heard. Headline numbers use a pre-registered input-side filter (21 formula-LaTeX ids + 1 broken recording removed; spoken formulas are destroyed by any TTS render before the system hears them). Heard accuracy climbs monotonically, $\mathbf{42.2}\% \rightarrow \mathbf{52.8}\% \rightarrow \mathbf{59.2}\% \rightarrow \mathbf{63.3}\%$ at 0/14/31/57% escalation, every tier's gain significant (paired bootstrap; Figure 12). A measured always-escalate arm closes the curve at 66.5% (P50 4.2 s): the last 43 points of escalation buy 3.2 points, because past the gated tiers the binding constraint is the transcription-uplink channel, not selection — the same sessions re-scored with gold-text expert answers reach 92.2%. The remaining gap to the channel-controlled counterfactual (escalated rows re-scored with gold-text expert answers) is dominated by *transcription-uplink* errors, not gate failure: the channel costs +2.3/+6.9/+12.8 points across tiers, and against the matched-rate random reference *internal to each view* the gate wins at every arm — in the deployed heard view by 7.2–9.5 points, in the channel-controlled view by 5.4–8.4 (Figure 12). Live arm accuracies on this pool carry a replication-noise floor of $\pm$2–3 points, so no claim here rests on a sub-3-point live delta. Full tables (both views, both pools), latency profiles, uplink diagnostics, and two boundary query classes (false-premise, real-time) are in Appendix G. A talker-on replay leaves the frozen probe intact (AUC 0.871/0.856 vs. 0.866 headless); TTS-on, first audible audio lands p50 0.55 s on local turns and 22–29 ms on escalated ones (the canned stall), expert content at p50 6.4 s with the stall covering the head of the wait (Appendix H, G). Re-run end to end in the *concurrent* regime (audio interleaved while the talker speaks, in-regime gate) the loop reproduces: monotone gains on all seven pools, speakable 44.5$\rightarrow$71.1%, selective still beating always-escalate (Table 12).

7 Frozen Transfer to Public Speech Benchmarks

The out-of-distribution test the type-shortcut audit demands: we take the deployed gate *frozen* — weights, features, and layer choice fit once on the internal calibration mix — and run the complete live loop of §6 on six public speech benchmarks: Speech TriviaQA, Speech Web Questions, Llama Questions, and Reasoning QA from OpenAudioBench [Li et al., 2025] (spoken renderings of Joshi et al., 2017, Berant et al., 2013, and the set of Nachmani et al., 2024), SD-QA (real dialectal human speech) [Faisal et al., 2021], and VoiceBench AlpacaEval [Chen et al., 2024, Taori et al., 2023]. All audio is the benchmarks' official pre-rendered audio; no query ever touches probe calibration.

Table 1: Main transfer results: methods × five public speech QA benchmarks (TriviaQA/WebQ/Llama Q. OAB official judge, n = 250; SD-QA/Reasoning zh reference-anchored, n = 200/202); heard accuracy in %, ±2–3 point replication floor. Bottom: second duplex family (NemotronLabs-VoiceChat-11B, same frozen recipe, offline re-mix, official judges, Appendix C); English-only, so its avg spans four benchmarks over its own floor, and its gate clears matched-rate random on every pool at every budget (per-budget random references and permutation tests: Table 7). Gold-inject view at matched budgets: Table 14. Δ = avg gain over always-local (points). AlpacaEval (1–5): 3.99→4.35 gated, 4.76 always-escalate, no gain over random on MiniCPM; the second family does gain (3.55→4.46 vs. random 4.23).

	TriviaQA	WebQ	Llama Q.	SD-QA	Reason. zh	avg	Δ
live heard view (deployed loop)							
always-local	66.4	57.2	84.0	51.5	58.9	63.6	—
+ gate @15%	73.2	62.8	90.4	60.0	65.3	70.3	+6.7
+ gate @30%	80.0	68.0	90.4	72.0	71.3	76.3	+12.7
+ gate @50%	**86.0**	**73.2**	**94.8**	**79.5**	**77.2**	**82.1**	**+18.5**
always-escalate (measured)	88.0	80.8	93.6	88.5	83.7	86.9	+23.3
always-expert (gold-text ceiling)	96.8	86.4	92.8	93.0	87.1	91.2	+27.6
second family: NemotronLabs-VoiceChat-11B (offline re-mix, official judges)							
always-local	35.6	37.6	70.5	31.0	—	43.7	—
+ gate @15%	49.8	48.0	78.6	42.5	—	54.7	+11.0
+ gate @30%	61.5	58.8	83.8	54.5	—	64.6	+21.0
+ gate @50%	**74.1**	**69.6**	**87.6**	**67.5**	—	**74.7**	**+31.0**
matched random @50%	65.6	60.2	81.8	60.0	—	66.9	+23.2
always-expert (ceiling)	95.5	82.8	93.2	89.0	—	90.1	+46.4
probe AUC (frozen)	0.789	0.785	0.806	0.792	0.683	0.771	—
probe AUC (2nd family)	0.775	0.771	0.720	0.754	—	0.755	—
official offline	75.5	70.2	—	—	—	—	—

The only per-pool adaptation is *label-free*: tier thresholds are quantiles of the probe's own score distribution on each pool, consuming no labels and no judge calls. It buys budget control, not discrimination: one absolute threshold frozen on the internal pool still beats matched-rate random on four of five pools, but fires at 2–48% across them (Appendix I). Where an official protocol exists we score with it (the OpenAudioBench judge; VoiceBench's 1–5 judge), and our chat-mode control reproduces the official anchors on-scale.

The frozen gate improves all five QA benchmarks (Table 1), monotonically in budget. On Speech TriviaQA the 9B system reaches **86.0% live** — against its own official offline 75.5% and 62.9% for Qwen3-Omni-30B [Xu et al., 2025b] at 3.3× the parameters: selective cloud escalation outperforms larger standalone backbones. The cleanest evidence that the signal is difficulty rather than surface is cross-lingual: an English-only calibration expansion moved *Chinese* Reasoning QA from AUC 0.621 to 0.683, the largest per-pool gain. On Llama Questions — single-type, so no shortcut to exploit — **selective escalation matches the expert at half the expert calls**: the probe's split is decisive (kept-local 97.6% vs. escalated 69.6%, z=6.46), the expert's whole advantage lands where the probe sends it (69.6%→88.8%, p<.0001), and a measured live always-escalate arm lands *below* selective (93.6 vs. 94.8 at half the calls); matched-rate precision reaches 95% of the base-rate cap (Appendix J).

Two honest negatives: **AlpacaEval** has no "missing fact" event to read, and **Web Questions and SD-QA are flat** (strict alias labels; the failures the gate ranks highest are ones the expert misses too). Neither is intrinsic — on the second duplex family the same recipe beats matched-rate random on *all five* pools, AlpacaEval included ($p \leq .0004$), so the flat pools are a headroom property of the pool × backbone pairing (Appendices C, J).

8 Discussion

What the signal detects. The pre-answer probe is a *retrieval-failure* detector: strongest where an empty lookup is observable, blind to false premises, and complementary to decode-time signals, which motivates a decode-time check behind it (Appendix G).

Limitations. (i) One phrasing per $p(\mathrm{True})$ variant; real-speech validation is scripted read speech in one dialect; formula-symbolic content is out of scope for the audio arm (any TTS render destroys it). (ii) The second-family replication covers mid-band readability and frozen-recipe transfer only — not matched-pair attribution, the live loop, or the concurrent regime. (iii) Labels come from one judge family (a stronger re-judge bounds deltas at $\leq$1 point), no human labels. (iv) One session per query per tier ($\pm$2–3 point floor); under concurrent prefill the frozen read degrades (AUC .87$\rightarrow$.76) and an in-regime refit recovers .82 — calibration must follow the regime (Appendix H). (v) Our live protocol is turn-based end-of-turn escalation on scripted single-turn queries; the harder setting — barge-in, disfluent self-correction, chained tool calls mid-utterance — is what full-duplex agent benchmarks now target [Lin et al., 2026], and porting the gate there is future work.

Conclusion. Duplex-trained speech models carry a competence signal their standard readouts miss — a mid-network read, available half a second before the first audible word. Never gone; only misread.

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A Model roster

Table 2: Model roster. “Pair” groups a fine-tune with its raw backbone control; the comparison statistic is always the within-pair difference.

model	type	pair	role
MiniCPM-o 4.5	omni + duplex FT	Qwen3-8B	system target
Qwen3-8B	raw	—	pair control
MiniCPM-o 2.6	duplex FT	Qwen2.5-7B	duplex replication
Qwen2.5-Omni-7B	streaming omni FT	Qwen2.5-7B	omni control
Qwen2.5-7B	raw	—	pair control
Qwen2-Audio-7B	audio FT	Qwen2-7B	audio-FT control
Freeze-Omni	frozen LLM + adapter	Qwen2-7B	adapter control
Qwen2-7B	raw	—	pair control
Mistral-7B-v0.3	raw	—	backbone diversity
GLM4-9B-chat	raw	—	backbone diversity

B Extended signal and readout analyses

Table 3: Signals on MiniCPM-o 4.5, judged failure labels. In-dist = calibration-split AUC (probes 5-fold out-of-fold); LOPO math = leave-one-pool-out AUC on held-out math; audio = the same readout under speech input (final/mid-layer probe: audio-input LOPO math / text-only calibration read on audio states). Full per-signal detail: Appendix B.

signal	in-dist	LOPO math	audio input
max entropy@4	0.696	≈chance on raw backbones	—
$p(\text{True})$-pre	0.807	trap 0.945 (probe: 0.232)	**collapses**
$p(\text{True})$-post (full draft)	**0.899**	—	degrades
final-layer probe	0.822	**0.372** (inverts)	0.936 (intact)
mid-layer (L22) probe	0.866	**0.931**	**0.855** (text-calibrated)

The in-distribution shortcut, quantified. The numbers behind §4.1’s claim that the final-layer probe’s in-distribution AUC of 0.822 is mostly pool identity: a linear read of the same hidden state classifies a query’s source pool at 95.8% accuracy; a pool-identity oracle alone reaches AUC 0.715; and appending true pool labels to the probe’s features moves it 0.822→0.821. In-distribution area over random is statistically on par between the gate and the pool oracle (Appendix E); the two separate only under LOPO, where the probe inverts to 0.372 and the oracle is undefined.

ROC curves. Figure 3 shows the calibration-split ROC for the signals of Table 3.

$p(\text{True})$ **backbone dependence.** Across backbones, post-draft $p(\text{True})$ beats the trained final-layer probe on four of five raw/duplex models tested; the exception (GLM4-9B-chat, 0.685 vs. probe 0.798) shows the pattern depends on the backbone’s self-evaluation quality.

Matched pairs. Table 4: with label coverage matched by subsampling the raw control to the duplex model’s per-pool fail rates, the raw backbone keeps LOPO math at 0.825/0.809 while duplex models sit at chance — label coverage is ruled out; the representation changed. Degradation is ordered raw $>$ omni-streaming $>$ duplex. Table 5 gives the by-depth summary behind Figure 2: the within-pair deficit at the best mid-layer is only -0.03 vs. -0.59 at the final layer; mean-pooled probes survive to the final layer on duplex models (LOPO math ≈0.80 at L35 on o4.5, 0.674 at L27 on o2.6 — 0.13–0.15 below each model’s mid-layer last-token peak, and uniformly weaker in-mix, -0.058 at L22); duplex damage is severe but local, omni-streaming damage mild but diffuse.

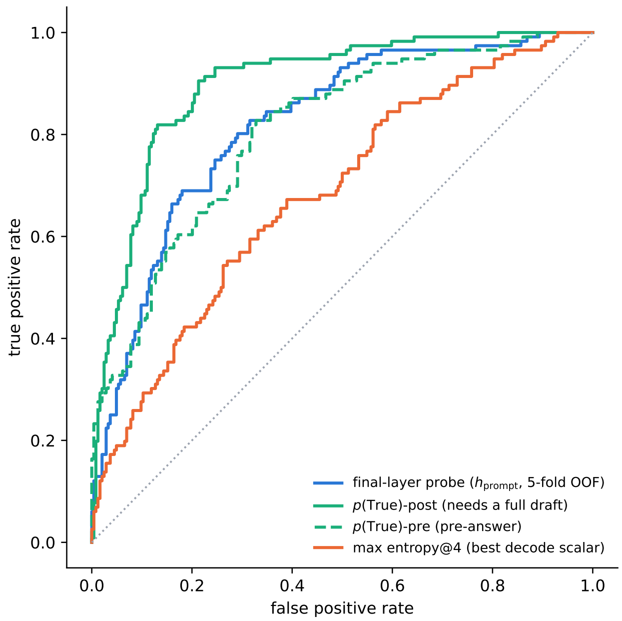


Figure 3: ROC on the calibration split. $p(\text{True})$-post leads but needs a full drafted answer; the probes and $p(\text{True})$-pre are pre-answer.

Table 4: LOPO transfer (AUC, held-out pool) of the final-layer last-token probe, Qwen2.5-7B family, label-coverage matched.

LOPO pool	Qwen2.5-7B (raw)	Qwen2.5-Omni	MiniCPM-o 2.6	[o4.5 vs Qwen3]
hard-math	**0.809**	0.745	**0.538** $\approx$ chance	0.372 (inverts)
hard-knowledge	0.680	0.613	**0.526** $\approx$ chance	0.61

Modality specificity of the cliff. Under audio input the final layer does *not* invert (L35 audio LOPO math 0.936 vs. 0.366 text); a speak-mode template control on text input leaves the cliff unchanged (0.362) — the operative variable is the modality of the input context, not the output mode.

Audio collapse of verbalized introspection. Table 6. Three controlled arms pin the mechanism on MiniCPM-o 4.5: *not perception* (ASR audit WER 0.074; $p(\text{True})$ on the model's own transcript snaps back to 0.074; the well-heard subset still collapses); *not a context prior* (irrelevant filler audio does not inflate p_{yes}); *binding* (duplicating the question as text alongside the audio fully restores introspection, 0.034). The verbal instance-check runs over text-token pathways that audio embeddings do not feed, while a linear probe reads the same information off the same forward pass at 0.93+. The failure implies its own cheap fix, confirmed twice: re-present the query as text before asking ("repeat-then-judge").

Real-speech validation. The matched-pair design rerun on 200 real human recordings (VoiceBench SD-QA, USA split) replicates all three audio-side findings: modality tax (0.400→0.450), audio overconfidence (paired p_{yes} shift $+0.089$; failure subset 0.415→0.581), and the layer structure (text-calibrated read on real-speech audio holds 0.76–0.80 at L22–L26, peak 0.800 at L25 — transfer across both content and modality).

Synthesis. End-to-end multimodal training *builds* the shared mid-layer core; duplex-style training additionally *damages* the readouts; omni-streaming gets both right; an adapter alone gets neither

Table 5: LOPO math transfer by depth (last-token probes).

model	best mid-layer	final layer	shape
Qwen3-8B (raw)	0.964 (L33/36)	0.958	plateau, no cliff
MiniCPM-o 4.5 (duplex)	**0.931 (L22/36)**	**0.366**	cliff, inverts
Qwen2.5-7B (raw)	0.893 (L21/28)	0.809	mild dip
Qwen2.5-Omni (streaming)	0.794 (L16/28)	0.746	diffuse depression
MiniCPM-o 2.6 (duplex)	**0.822 (L21/28)**	**0.540**	cliff

Table 6: Pre-answer introspection under audio input. Trap p_{yes} = mean stated probability of answering correctly on a 100%-fail pool (lower is better).

model	FT type	trap p_{yes}: text $\rightarrow$ audio	audio $p(\text{True})$-pre AUC
MiniCPM-o 4.5	duplex	0.055 $\rightarrow$ **0.556** (collapse)	0.786 (trap dead)
MiniCPM-o 2.6	duplex	0.196 $\rightarrow$ 0.345	**0.491** $\approx$ chance
Qwen2.5-Omni	omni-streaming	0.279 $\rightarrow$ 0.213 (intact)	0.727
Freeze-Omni	frozen + adapter	0.165 $\rightarrow$ 0.112 (intact)	0.612 (capability caveat)

(probe readout text-fragile, verbal readout audio-fragile; in both cases knowledge survives and a readout breaks, and in both cases the breakage is duplex-fine-tune-specific).

The deployed probe. The mechanistic probe is a single-position L22 read. The deployed probe adds two zero-latency pooled positions (an 8-token tail mean and the running user-audio mean), refit on a calibration mix widened to 2,310 public-benchmark queries disjoint from every evaluation benchmark; a pre-registered guard held (frozen internal test AUC 0.860→0.879) and feature selection never saw the externals. In the mid-layer band the in-mix tradeoff optimum is broad (L20–L30) and the mid-layer probe's in-distribution area advantage is modest — its real advantage is LOPO robustness, which an in-mix test cannot show.

C Second duplex family: NemotronLabs-VoiceChat-11B

The pre-registered prediction test: with the identical recipe (same audio, same judge, 600-query calibration), the layer sweep on this hybrid-Mamba duplex model (56 blocks: 27 Mamba-2 / 4 attention) peaks mid-band at L30–34 (eot-last AUC 0.714 vs. 0.693/0.682 at the endpoints); stacking the same three positions lifts OOF 0.714→0.761→0.790; the frozen-methodology probe transfers to the external pools at AUC 0.781/0.793/0.754/0.701 — the deployed-probe band reached with a quarter of the calibration data. Boundary notes: the backbone is much weaker at knowledge retrieval (striviaqa local 0.32 vs. MiniCPM 0.62, same judge) and English-only (the cross-lingual pool has no analog). The live-loop port and the trained-checkpoint ablation remain future work; the port is gated by the released inference path, not by the architecture. The only supported loading path (the NeMo Speech `nemotron-labs-voicechat` branch) runs the duplex loop *cacheless* — every 80 ms frame re-runs the full prefix, an $O(T^2)$ schedule that cannot hold the frame clock in real time — although the hybrid backbone's Mamba-2 state is exactly the $O(1)$ per-frame recurrence a streaming port would need; wiring stateful inference into the released duplex frame protocol is the engineering gap. (Offline, the same property is a convenience: the final frame's forward contains every position, so one hook captures the whole eot window.) Accuracy-side conclusions do not wait on the port: the re-mix arithmetic below tracks measured live arms at 0.011 mean deviation on MiniCPM, inside the $\pm$0.02–0.03 replication floor.

Escalation re-mix on every transfer pool. Completing the grid of Table 1 on this family: an offline re-mix (the same top-r reconstruction arithmetic that, on MiniCPM, tracks the measured live arms at 0.011 mean deviation) sends the top-r of each pool by NVDA probe score to the measured GPT-5.5 outcome and keeps NVDA's own answer for the rest, scored with each pool's official protocol (the OAB judge for the three OpenAudioBench pools — NVDA local floors 0.352/0.376/0.705 —

our judge for SD-QA, VoiceBench 1–5 for AlpacaEval; the expert answers the heard transcript and is judged directly, so the relay-back channel is excluded — including the measured relay outcomes where they exist shifts tier points by $\leq$0.032). **Selective escalation beats matched-rate random on all five pools** (permutation $p \leq 0.0004$ at the 30% and 50% budgets, $p < 0.03$ at 15%; Table 7, Figure 4) — including Web Questions, SD-QA and AlpacaEval, the three pools that are flat or negative on MiniCPM (§7). Selectivity is nearly unchanged (AUC 0.72–0.78 under official labels); what changed is headroom — floors of 0.31–0.38 against MiniCPM's 0.51–0.66 — and on AlpacaEval the failure species: the terse English-only model fails open-ended queries *hard* (the probe-selected half scores 3.01 vs. 4.08 kept), where MiniCPM's open-ended failures are soft quality gradations a wrongness detector cannot see. The negatives are properties of the pool × backbone pairing, not of the signal.

Table 7: The frozen-recipe NVDA probe re-mixed at fixed escalation rates, official judge per pool, each against its own matched-rate random reference (same measured outcomes, escalation set drawn at random; identical at 0% and 100% by construction). AlpacaEval rows are VoiceBench 1–5 scores; others accuracy. Every pool clears random at all three budgets: $p \leq 0.0004$ at 30% and 50%, $p < 0.03$ at 15% (20k permutations, `scripts/18_nvda_random_check.py`).

pool	judge	n	0%	15%	30%	50%	100%
Speech TriviaQA	OAB	247	0.356	0.498	0.615	0.741	0.955
matched random				0.446	0.536	0.656	
Speech Web Questions	OAB	250	0.376	0.480	0.588	0.696	0.828
matched random				0.444	0.512	0.602	
Llama Questions	OAB	234	0.705	0.786	0.838	0.876	0.932
matched random				0.739	0.773	0.818	
SD-QA	ours	200	0.310	0.425	0.545	0.675	0.890
matched random				0.397	0.484	0.600	
AlpacaEval (1–5)	VB	199	3.55	3.83	4.13	4.46	4.92
matched random				3.75	3.96	4.23	

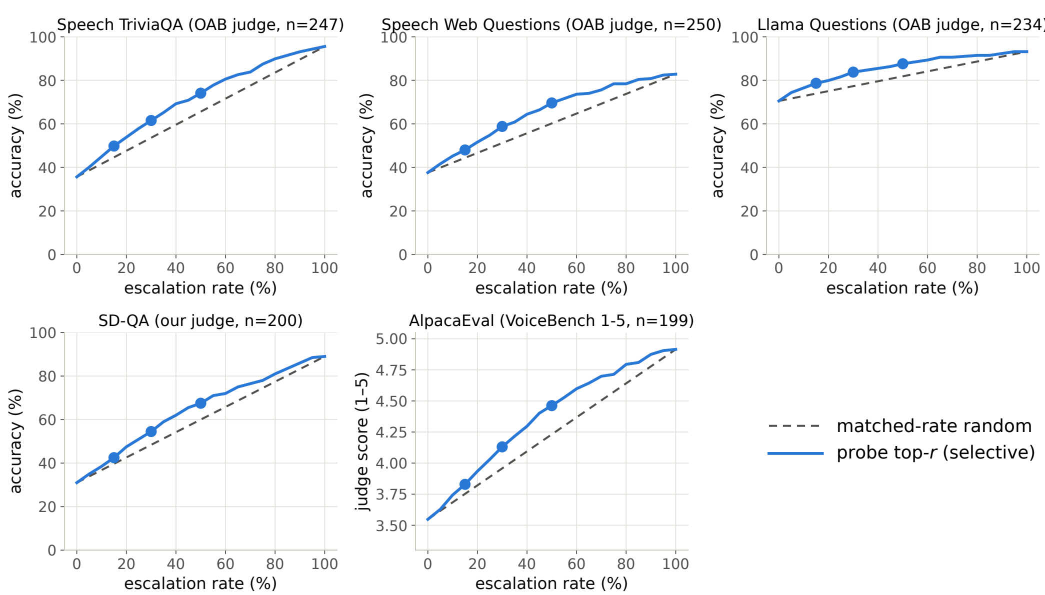
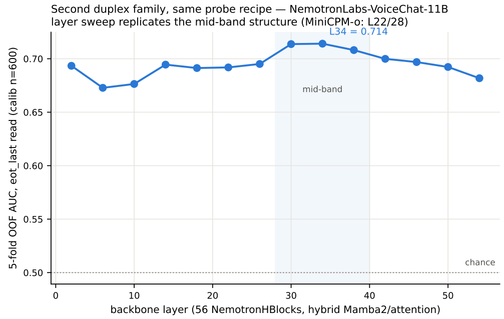
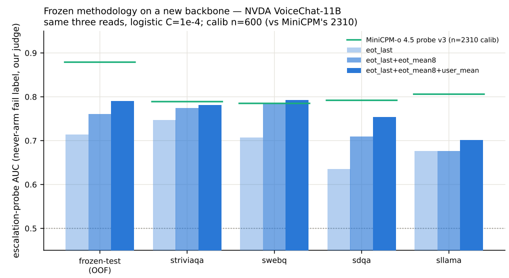


Figure 4: Re-mix curves for the frozen-recipe NVDA probe: selective (top-r by probe score) vs. matched-rate random on the five transfer pools, official judge per pool. Every pool separates, including the three that are flat on MiniCPM (all five pools, $p \leq 0.0004$ at 30/50% and $p < 0.03$ at 15%, permutation).



(a) layer sweep



(b) external transfer

Figure 5: Second-family test: (a) the usable read again peaks mid-network; (b) frozen-recipe transfer lands in the deployed-probe AUC band.

D What breaks at the output: mechanism audit

§4.2 reports *that* the final-layer last-token read inverts. This section reports what we could and could not establish about *why*. Everything below uses the Phase-5d all-layer prefill dumps (every decoder layer, last-token and mean-pooled, same 600 queries) and the identical probe protocol as the layer sweep: calibration split, L_2 logistic regression, LOPO trained on the four non-math pools and scored on held-out math. The reproduction check recovers the published L22 and final-layer numbers to $\leq$0.012 (`scripts/14-17_*.py`).

Four mechanisms ruled out. Table 8 lists them. (i) *Speak mode.* Prefilling the same text queries under the TTS/speak-mode template leaves the cliff exactly where it was (final-layer LOPO math 0.362 vs. 0.366 plain), so the readout is not being displaced by a prepare-to-speak state. (ii) *Modality axis.* The mean audio-minus-text direction at the final layer is orthogonal to both the inverting final-layer probe and the transferring L22 probe (cosine 0.013 and 0.027; chance is $1/\sqrt{4096} = 0.016$), so the cliff is not the audio/text split that §4.3 measures. (iii) *More shortcut available late.* Source-pool identity is linearly decodable at 0.88–0.96 at *every* depth, in the duplex model and its raw backbone alike — the query-type shortcut of §4.1 is not something the late layers add; it is what remains once the competence signal stops being readable. (iv) *Wholesale representational drift.* Raw linear CKA between the duplex model and its backbone appears to collapse late (0.80→0.47), but transformer residual streams carry a few massive-activation coordinates that dominate the Gram matrix; with per-feature standardization the same comparison is nearly flat (0.82→0.78) and does not localize the cliff. Projecting out the massive-activation axes, or the pool-mean directions, rescues nothing (0.35–0.40 at every $k \leq 20$, against a random-direction control of 0.37).

Table 8: Mechanism candidates for the final-layer inversion. Final-layer LOPO hard-math AUC on MiniCPM-o 4.5 unless noted; 0.366 is the unmodified baseline and 0.931 the L22 read.

candidate	test	verdict
prepare-to-speak state	speak-mode template prefill	0.362, no change
modality (audio vs. text) axis	cosine with the probe direction	0.013 $\approx$ chance
late layers encode more pool identity	pool decoding by depth	0.88–0.96 at all depths
representation rewritten late	standardized CKA vs. backbone	0.82→0.78, flat
massive-activation coordinates	project out top-k such axes	0.35–0.40, no rescue
query-type directions	project out pool-mean directions	0.34–0.45, no rescue
delivery to the output position	**residual component of the read**	**0.93→0.27 (Fig. 6)**

What does separate the models. Figure 6 measures the probe rather than operating on the model. For the probe trained at each depth on the four non-math pools, we split its held-out score into the part carried by that layer's five dominant principal directions (estimated from training rows only) and the residual, $s = w^\top Px + w^\top(I-P)x$. The residual component is the distributed read, and it is the one that transfers: it holds 0.85–0.93 through mid-network on MiniCPM-o 4.5 and then decays to **0.27** at the output, with the same decay on MiniCPM-o 2.6 (0.81→0.52) and *no* decay on either raw backbone (Qwen3-8B 0.86, Qwen2.5-7B 0.71 at their final layers) or on the omni-streaming control (Qwen2.5-Omni holds 0.75 to its output with a flat variance share) — the decay tracks duplex fine-tuning specifically, not streaming or audio fine-tuning in general. On MiniCPM-o 4.5 the score also concentrates: the dominant subspace carries 39% of the held-out score variance at the final layer against $\leq$0.18 at every earlier depth and 0.05 for its backbone; on MiniCPM-o 2.6 this concentration is present but weak (0.16 vs. 0.14 for its backbone), so we report it as a property of the stronger duplex model, not a general law.

Direct turn-control tests. Two experiments target the turn-control story head-on rather than by exclusion (`modal_interp.py`, `scripts/19_*.py`). *Control-token logit lens:* applying each model's own final norm and unembedding to the stored per-layer states, the probability mass on the trained-in control vocabulary (added/special tokens) *rises* over MiniCPM-o 4.5's last layers (log-mass -23.8 at L32 to -11.8 at the output) while the raw backbone's falls by fifteen log-units over the same span — the duplex output layer, uniquely, stays ready to emit control tokens. But the fine-grained prediction fails: the inverting probe direction is no more aligned with the control tokens' unembedding rows than with random vocabulary rows (max $|\cos|$ 0.069 vs. 0.068), and per-query control mass correlates with the probe score at only r=0.18. *Listen/speak intervention:* re-prefilling 60 calibration queries with a still-listening vs. answer-now suffix moves the final-layer state substantially in *both* models (relative displacement 0.27 duplex, 0.59 raw — cue text is just prompt semantics), and the cue direction is orthogonal to the inverting probe ($|\cos|$=0.019, chance 0.016), shifting its score by only $+0.23\,$sd (and the deployed L22 read by $+0.08\,$sd). Together: the duplex output layer is measurably control-biased in its output distribution, but the inversion is not carried by control-token directions or by an explicit listen/speak axis — the simple takeover story fails its own direct tests.

What we do not claim. Destructive tests do not settle the mechanism. Removing the top five principal directions from the final layer does raise LOPO math from 0.378 to 0.758 — but the identical operation costs the raw backbone 0.90→0.14 and costs the intact L22 read 0.931→0.760, so it is a damaging intervention that happens to help a broken readout, not evidence of a duplex-added subspace. We therefore state the finding at the level the evidence supports: duplex fine-tuning does not erase the competence signal, it stops delivering it in distributed form to the final last-token position, and the layer where delivery fails is also the one whose output distribution stays control-biased. Attributing the loss to a specific duplex objective needs training-time access we do not have; the practical consequence — read mid-network — does not depend on the attribution.

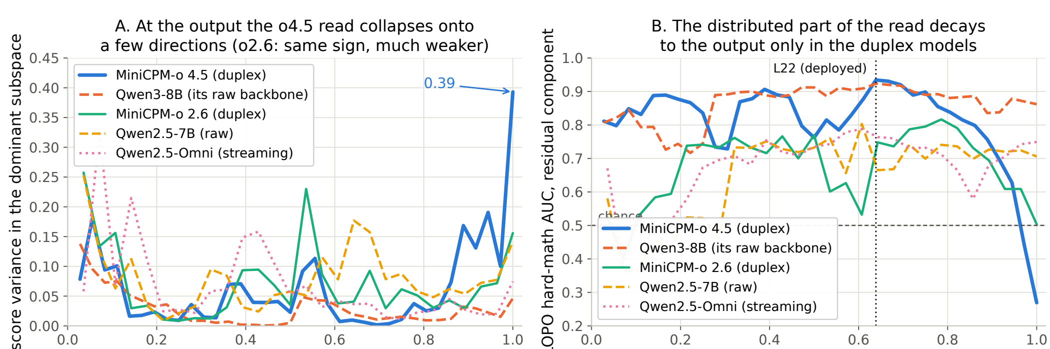


Figure 6: Non-destructive decomposition of the probe's held-out score at every depth. **Left:** share of score variance carried by the layer's five dominant directions. **Right:** transfer AUC of the residual (distributed) component — intact to the output on both raw backbones and on the omni-streaming control, decaying to inversion over the final layers of both duplex models.

E Query-surface router audit

A RouteLLM-style TF-IDF→LR router under identical labels reaches OOF AUC 0.669 (below the pool oracle's 0.715); a 24-configuration sweep tops out at 0.689, so this is the query surface's ceiling, not tuning. Under LOPO it collapses to 0.38–0.57 and scores the 100%-fail trap pool at 0.230 — it misses the pool a router most needs. The training receipt: at the argmax threshold its out-of-fold accuracy exactly matches the majority class; the internal probes under the same accounting clear majority by 9–29 points (Figure 8). Nor is it data-starved: the identical recipe on RouterBench [Hu et al., 2024] (36,497 prompts) reaches only in-domain AUC 0.710 with leave-one-benchmark-out at chance; RouteLLM's released preference-trained routers score 0.523/0.533 zero-shot on our pool and rank the trap pool below ordinary hard pools; our router transfers to RouterBench below chance (0.440). On audio labels the router improves (test 0.814) but still trails the mid-layer probe (0.879) and cannot fire mid-stream. In distribution, the gate's area over random ($+0.054$) is statistically on par with the pool oracle's ($+0.042$; paired bootstrap delta CI $[-0.003, +0.027]$, p=0.13) — the gate's case is transfer, not the in-distribution margin. $p(\mathrm{True})$ tradeoff variants: Figure 9.

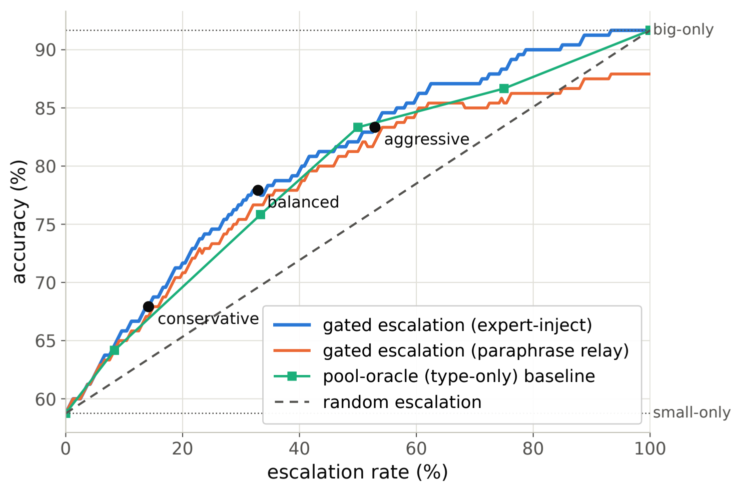


Figure 7: Accuracy vs. escalation rate, frozen test split: gated escalation vs. random-escalation and pool-oracle (type-only) baselines between the small-only and big-only endpoints.

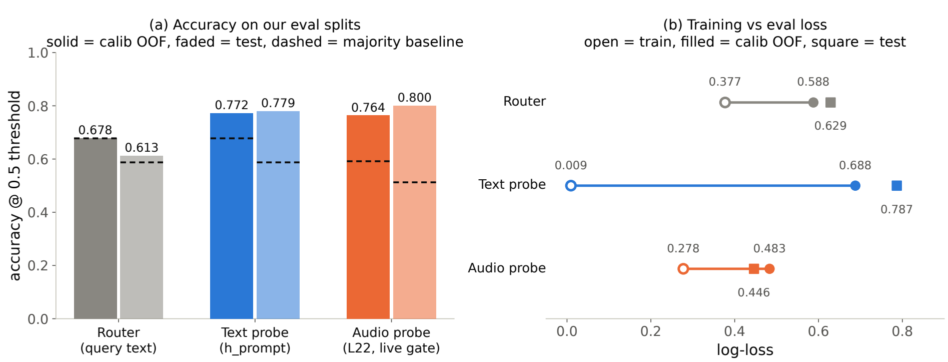


Figure 8: Identical training receipt for the external router and the two internal probes: the router never clears the majority-class baseline; both probes do, by 9–29 points.

F Gate timing details

Against the model's own thinking mode. MiniCPM-o 4.5 ships a built-in reasoning mode — self-escalation in place. Gated-cloud escalation dominates all-THINK on both accuracy and latency

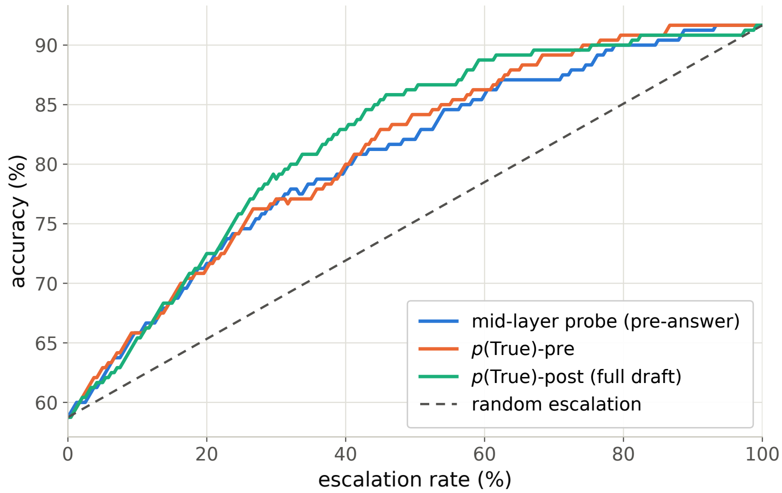


Figure 9: Offline tradeoff for the $p(\text{True})$ signals (pre-answer and post-draft).

at every escalation rate (gated-cloud@33%: 78.7% at mean 6.5 s vs. all-THINK 63.7% at 22.4 s; Table 9). The reason is mechanistic: the gate flags knowledge and trap failures, which extra compute cannot fix — so external escalation is necessary, not merely better.

Table 9: Deployment policies, frozen test split (n=240); latencies measured.

policy	acc (%)	lat. mean (s)	P50	P95
fast-only	58.8	3.0	3.5	4.2
all-THINK (built-in)	63.7	22.4	17.2	60.1
gated-think @33%	61.3	12.1	3.6	47.6
gated-cloud @15%	68.8	5.3	3.6	10.1
gated-cloud @33%	**78.7**	**6.5**	3.6	20.8
gated-cloud @50%	85.8	7.0	3.6	24.4

Measurement protocol. Gate latency is wall-clock from prefill start to the L22 score being available (truncated forward + probe dot product), CUDA-synchronized, one H100, 50 queries per modality, three warmup calls excluded, interleaved per query with the reference arms; P50/P95: gate 20/25 ms text and 45/104 ms audio vs. full-prefill TTFT 36/47 and 68/144 ms. Network and expert-RPC initiation are excluded: the cloud call fires asynchronously at the gate, and its user-facing cost is reported as expert-content-audible latency in Appendix G. Absolute prefill wall-times vary with call overhead across benches; the robust cross-bench statistic is the ratio — the L22 state is ready at ∼57–61% of prefill.

Per-layer truncated-forward cost vs. quality (Figure 10): the L22 hidden state is available at ∼57–61% of prefill wall-time while L22 is simultaneously the in-mix quality peak; forking much earlier is both weaker and modality-specific, so the fork belongs at ∼55–60% depth. If the cloud call launches at the gate, the expert result arrives before the local answer finishes for 40–58% of the whole mix but only ∼20–31% of *escalated* queries (they skew to short local answers); the residual wait is P50 2–3 s, P90 ∼27 s (Figure 11) — the stall-phrase budget the injection protocol must cover.

G Live-loop details

What the signal detects. The pre-answer probe is specifically a *retrieval-failure* detector, strongest on missing-knowledge and long-tail-trap failures where the empty lookup is observable in the model's state. Execution failures (stuck mid-derivation) surface in *decode-time* signals instead, and metacognitive failures — false-premise questions, where no failure event occurs at all —

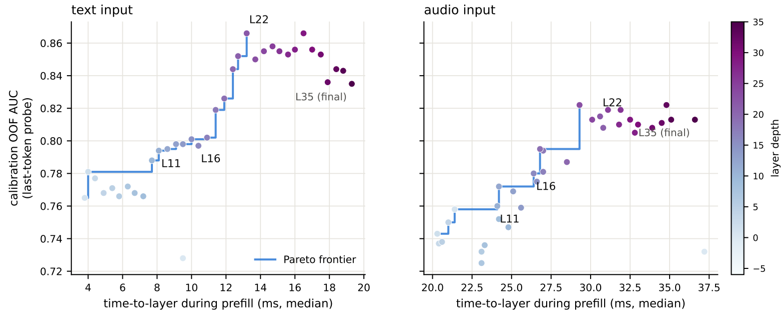


Figure 10: Fork-depth Pareto: per-layer wall-time vs. gate quality.

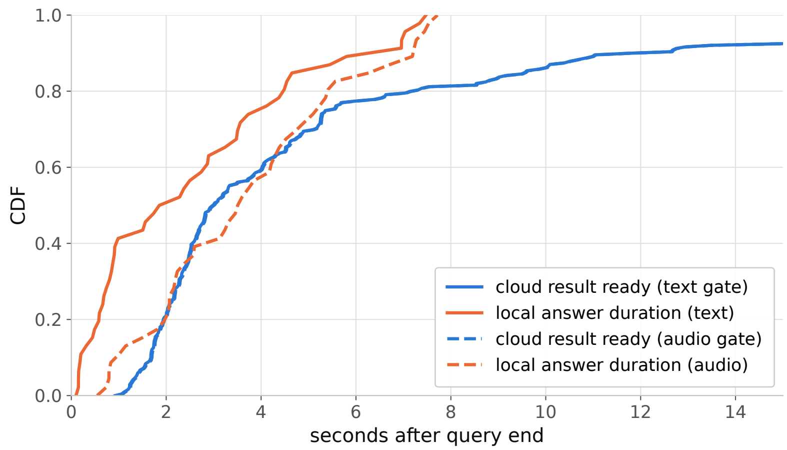


Figure 11: Cloud round-trip vs. local answer duration: probability the expert result arrives before the local answer ends.

are invisible to every signal we tested, query-surface routers included (fire rate 18% vs. trap's 90%; Appendix G), which motivates a decode-time check behind the pre-answer gate.

Full live tables. Table 10 (both scoring views, speakable subset and full pool) and Table 11 (measured response latency). The conservative tier *shortens* the P95 tail: its few escalations displace exactly the longest local decodes. Figure 13 joins accuracy and latency; Figure 14 replays three real sessions from recorded timers.

Time to first audible audio. The sweep above is text-out; a TTS-on re-run of the balanced arm (same 240 queries, talker head + vocoder on, stall pre-synthesized once in the talker's own voice) measures speech onset directly. The gate decision is unchanged — the TTS token enters the context only after the end-of-turn read (read p50 21 ms / p99 32 ms), and the same 84/240 queries fire. From gate decision to first audible audio: **local answers p50 0.552 s / p95 0.599 / p99 0.644**; **escalated turns p50 0.022 s / p99 0.029** — the canned stall (3.2–4.1 s of speech) plays the moment the gate fires, so the escalated path reaches first audio 25× faster. Expert *content* becomes audible at p50 6.4 s / p95 26.0 / p99 34.0 (cache-corrected expert round-trip + first relay PCM, p50 0.66 s after the answer returns); per-turn dead air between stall end and expert content is p50 2.5 s / p95 22.4 / p99 30.7, with 25/84 escalated turns fully covered — the stall covers the head of that wait, and Table 11's completion profile bounds the tail.

Speculative prefetch, measured and dropped. Framed as speculative escalation (draft–verify at sentence granularity), its acceptance rate — does an expert answer to the partial transcript answer the full question — is 0.07/0.19/0.51 at 25/50/75% of the audio; at the mid-stream gate's typical fire

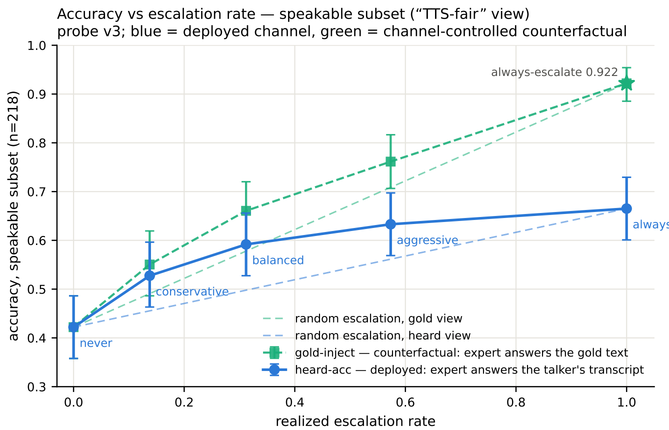


Figure 12: Live dual-view tradeoff (n=218 speakable subset). Blue: honest heard accuracy of the live system (paired-bootstrap CIs). Green: the channel-controlled counterfactual — same sessions, escalated rows re-scored with gold-text expert answers. Each view is plotted against *its own* matched-rate random reference (faint dashed, same floor, each view's own 100% endpoint): selection beats random within both views at every arm, by 0.072–0.095 heard and 0.054–0.084 gold. The vertical view-to-view gap is the speech-channel price; the floor-to-floor gap to offline operation is the audio + streaming-answer tax.

Table 10: Live streaming sweep, paired bootstrap 95% CIs (B=10^4). Top: speakable subset (n=218); bottom: full pool (n=240). Gold-inject = escalated rows re-scored with gold-text expert answers; channel cost = paired difference. * CI excludes 0; † CI lower bound at zero.

tier	esc	heard-acc	Δ vs. floor	gold-inject	channel cost
never	0%	0.422 [0.36,0.49]	—	0.422	—
conservative	14%	0.528 [0.46,0.59]	+0.106*	0.550	+0.023†
balanced	31%	0.592 [0.53,0.66]	+0.170*	0.661	+0.069*
aggressive	57%	0.633 [0.57,0.70]	+0.211*	0.761	+0.128*
always (synth)	100%	—	—	0.922 [0.89,0.95]	—
never (full)	0%	0.383 [0.32,0.45]	—	0.383	—
conservative	16%	0.483 [0.42,0.55]	+0.100*	0.525	+0.042*
balanced	35%	0.554 [0.49,0.62]	+0.171*	0.654	+0.100*
aggressive	61%	0.596 [0.53,0.66]	+0.212*	0.771	+0.175*
always (synth)	100%	—	—	0.917 [0.88,0.95]	—

point 6–8 of 10 calls would be wasted. The trap pool's floor (0.00/0.10/0.27) shares the back-loaded-core property that breaks mid-stream probe reads.

Conflicting-injection controls (172 sessions). Fabricated conflicting expert answers by within-pool derangement, four framings: worst-case comply is 0.00 neutral / 0.53 deployed authority / 0.79 strong-override, while *seeding* words into the talker's mouth backfires (comply 0.25, lip-service 0.62 — forced speech is treated as revisable). Math is silently re-derived rather than relayed or disputed. Model-wrong × wrong-injection swallow rate is 0.64: expert quality is the accuracy ceiling of the cascade.

Uplink diagnostics. On the escalated population the expert answering the talker's own transcript scores 0.585 vs. 0.871 on gold text. Prompt-level rescues on the self-transcript are no-ops (the damaging errors are internally-coherent entity substitutions); replacing the ears helps: open-weights Whisper recovers little (+4 pp) but a stronger cloud ASR on the same audio recovers 38% of the gap (0.585→0.694, McNemar p=0.007) — diagnostic only (its critical-path latency and external-pool survival are unmeasured). A same-model gold-text control with an audio-capable expert shows the

Table 11: Measured response latency (s), n=240 per arm (latency is channel-independent). P99 on n=240 is indicative only.

tier	mean	P50	P95	P99	Δmean	ΔP50	ΔP95	ΔP99
never	4.6	2.0	15.8	17.8	—	—	—	—
conservative	4.6	2.6	12.8	23.0	+0.0	+0.6	−3.0	+5.2
balanced	5.8	3.5	15.9	32.7	+1.2	+1.5	+0.1	+14.9
aggressive	6.6	4.4	18.5	33.0	+2.0	+2.4	+2.7	+15.2

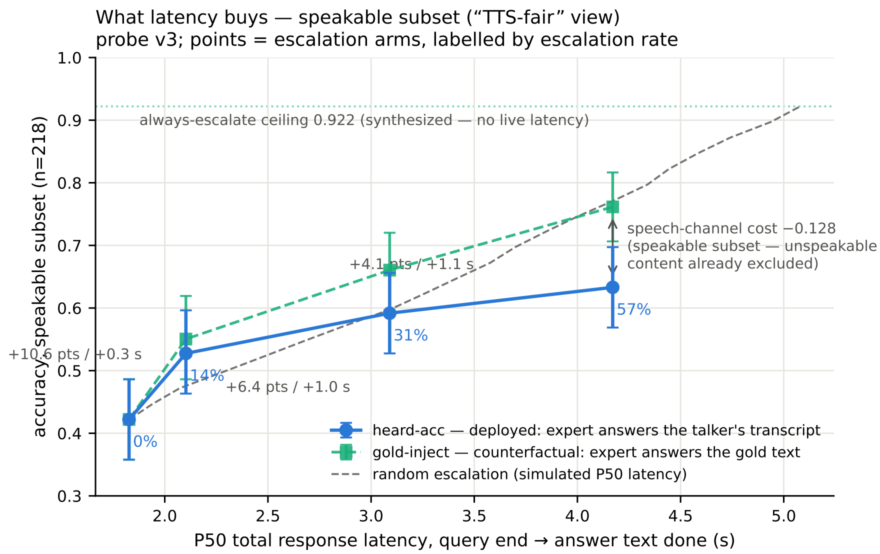


Figure 13: Accuracy vs. per-arm P50 latency (speakable subset), both scoring views, with the simulated-P50 random reference.

audio channel itself is lossless for speakable content: the leak is the talker's transcription, not the synthesis or the relay (relay drops: 5 cases of 108; corrupted transcripts dominate the escalated heard-failures, first-generation sweep).

Boundary query classes. *False premises*: the model fails substantially (0.63 text / 0.47 audio) and the expert would help (0.80), but the gate is blind (fire@balanced 0.18 vs. trap's 0.90; no score separation) — answering along a false premise does not feel like inability. *Real-time queries* ("what is NVDA trading at"): the gate is not blind, the class was simply absent from every static training family; extending calibration with FreshQA [Vu et al., 2023] (fast-changing labeled escalate a priori, never-changing as in-family controls) raises held-out fast-changing fire rates to 0.80/0.93/1.0 across tiers with frozen-test AUC unchanged (0.877 vs. 0.879). One calibration subtlety: tier budgets must remain quantiles of the deployment-mix scores, or the new capability is spent on its own threshold shift.

Demonstration system. The pipeline runs as an interactive web demo on one H100: live microphone audio, end-of-turn read (∼54 ms) against the balanced threshold, escalated turns stall, consult the expert, and *speak* the relayed answer through the model's native TTS head (probe context and thresholds untouched). Two demo-only extensions: a web-search tool for the expert (real-time queries then return live values), and backchannel-aware barge-in — speech over the talker ducks playback, a fast ASR pass returns the floor on backchannels ("ok", "mm-hm"; an English+Chinese lexicon) or commits the interrupt, aborting generation at chunk granularity and seeding the next turn. All measured arms keep the tool-free expert and turn-based loop; the demo will be made available upon publication.

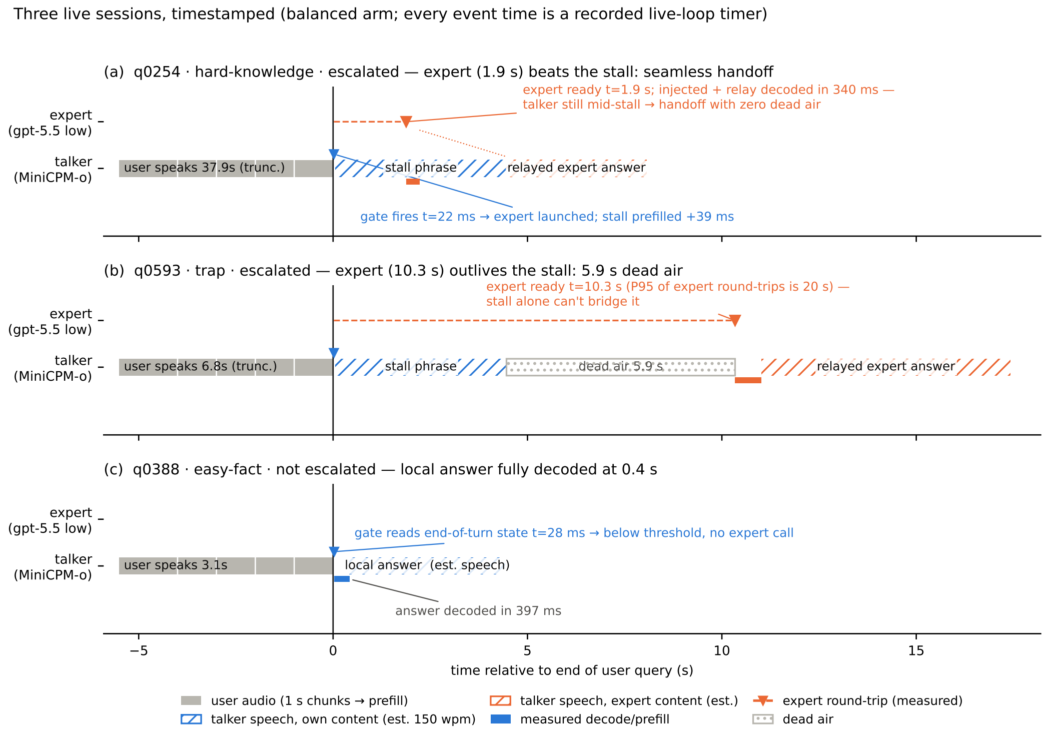


Figure 14: Three real sessions from recorded timers: expert beats the stall phrase (seamless); expert outlives it (dead air, the P99 signature); gate stays closed (local answer fully decoded 0.4 s after end of turn). Speech occupancy is estimated at 150 wpm (hatched); the measured arms are text-out, so no vocoder onset is plotted.

H Duplex-validation sweep: the gate under the talker

The headless live sweep (§6) never engages the talker head; this sweep replays its full 240-query pool through the deployed voice loop above to test whether the frozen probe read survives the spoken regime. Two arms, one session per query, deployed v4 gate with scores logged but escalation disabled (the arm isolates gate validity, not routing outcomes). **Clean**: the query streams as mic-shaped PCM (pool TTS audio plus steady noise at RMS 0.008), the energy VAD ends the turn, the talker answers aloud, and the session is cut after the first audible chunk. **Overlap**: an easy warmup question (probe score <0.25, locally answered, >5 s spoken answer) puts the talker mid-answer, and the target query is then spoken over it; sustained overlapping speech commits a barge-in whose audio seeds the next turn. The barge-in path engaged on 237/239 pairs.

Against the frozen local-failure labels of the same pool, the identical frozen probe reads: headless AUC 0.866 $[0.817, 0.909]$, clean 0.871 $[0.824, 0.914]$, overlap 0.856 $[0.803, 0.901]$; barged-only subset (n=237) 0.854. Paired bootstrap deltas vs. headless are $+0.005$ $[-0.025, +0.035]$ (clean) and -0.009 $[-0.036, +0.016]$ (overlap) — both inside the ±0.02–0.03 live replication floor. Per-query scores correlate across regimes at r=0.90–0.92, and the balanced threshold makes the same fire/hold decision as the headless read on 0.887 (clean) and 0.912 (overlap) of queries. The end-of-turn read costs 24–26 ms at p50 (p95 $\leq$30 ms) in both arms, with the first audible answer chunk arriving $\geq$0.5 s after the gate decision (p50 $\approx$0.58 s), so the decision precedes any audible commitment. Scope: this loop keeps per-turn session control — the mic stays open while the talker speaks and interruptions seed the next read, but incoming audio is not prefilled *during* generation. The concurrent regime is measured separately below.

Concurrent prefill: interleaved listen/speak in one KV stream. A third arm drives the streaming API in time division: while the talker is mid-answer, each 1 s chunk of the target query is prefilled into the *same* session between generation chunks, so the end-of-turn read fires while generation is still active (238/240 queries, full interleave coverage). The carrier answer is teacher-forced: under sampling the duplex model reliably yields the turn — it emits EOS within $\sim$2 s of user audio appearing in its context, the trained turn-taking behavior surfacing under interleave. This is the one shift on the transfer ladder that breaks the frozen read: scores drift up (mean 0.48→0.71; the frozen balanced threshold's fire rate jumps 37→75%), AUC drops 0.869→0.758 [0.693, 0.816] (paired ΔAUC -0.109 [-0.158, -0.064]), and label-free requantiling recovers decision agreement only to 75%. The information, however, survives in the state: refitting the *same* 12288-d linear read on 360 calibration queries collected in-regime reaches **0.818** [0.761, 0.871] on the held-out 240 (calibration OOF 0.775) — more than half the gap recovered at a quarter of the deployed probe's calibration scale. The mid-layer competence direction persists under concurrent listen/speak; unlike every earlier shift (modality, external pools, talker-on), calibration must follow the regime. Run-to-run replication of the concurrent read correlates at 0.895. Caveat: the interleave goes through the public streaming API — turn-role headers enter mid-generation and the carrier text is forced — so this measures the concurrent *state*, not the official duplex serving format.

The full loop in the concurrent regime. With the in-regime gate (thresholds = label-free quantiles of each pool's own concurrent never-arm scores), the complete escalation loop — carrier speaking, target audio interleaved, EOT read mid-generation, stall → expert → relay — was re-run end to end on all seven pools (Table 12). The mechanism reproduces: heard accuracy rises monotonically in budget on every pool, realized rates track nominal (10–51%), and *selective escalation beats always-escalate* on the internal pool (66.2 vs. 61.7%; speakable subset 71.1 vs. 66.5%) and on Llama Questions (86.4 vs. 85.6%, our judge) — the turn-based signature result carries over. On the speakable subset the concurrent loop reaches **44.5→71.1%** at 63% escalation (turn-based reference 42.2→63.3% at 57%). Floors move by pool — the regime cost story: Llama ≈ unchanged, TriviaQA -16, Reasoning zh -18 (the English carrier hurts the zh pool most), WebQ $+4.8$ under the official judge (within the $\pm$2–3 replication floor; the larger our-judge shift is alias-matching style sensitivity). AlpacaEval reverses its turn-based honest-negative: 4.36→4.60 (1–5 scale, always 4.76). Two honest notes: the in-regime probe, calibrated on 360 internal rows only, transfers externally at AUC .57–.67 against the full-scale turn-based probe's .68–.81 — a calibration-*scale* gap, not a signal gap (internally, where calibration matches the distribution, it reads .857 vs. the turn-based .879) — and the internal conservative threshold under-fired (2.5% realized vs. 15.8 nominal; the top calib quantile did not transfer, balanced and aggressive did). Latency holds: EOT read p50 23 ms, canned-stall onset 27–34 ms, local first audio p50 .655 s (vs. .552 turn-based — the price of the longer concurrent context).

Table 12: The full escalation loop in the concurrent regime, all pools, judge-matched to Table 1 (OAB official for TriviaQA/WebQ/Llama; ours for SD-QA/Reasoning/internal; VoiceBench 1–5 for AlpacaEval). Gate = in-regime probe (360-row calibration), per-pool label-free quantile thresholds. Accuracies in %; AUC = in-regime probe vs. never-arm failure, decimals.

pool	never	@15%	@30%	@50%	always	AUC
TriviaQA	50.4	54.0	64.0	71.6	91.6	0.636
WebQ	61.6	65.2	68.8	71.2	81.6	0.636
Llama Q.	81.6	83.2	84.8	90.8	91.2	0.672
SD-QA	49.5	55.5	60.5	70.5	89.5	0.636
Reason. zh	40.6	48.5	58.4	64.4	80.7	0.570
internal (240)	40.4	40.0	52.1	66.2	61.7	—
internal speakable (218)	44.5	43.6	56.0	**71.1**	66.5	0.857
AlpacaEval (1–5)	4.36	4.45	4.53	4.60	4.76	—

A serendipitous recovery behavior. An accidental run surfaced one bonus behavior worth reporting: when a long question contains a $>$1.25 s internal pause, the endpointer fires early and the talker starts answering the fragment — but the question's own continuation then triggers barge-in, and the remainder becomes the next turn, where the gate simply re-scores it. Early end-pointing thus degrades into an extra turn rather than a lost query — the honest duplex-ish recovery available to a turn-based loop.

I Fixed-threshold transfer and the expert-benefit oracle

Both analyses below run in the *gold-inject* view (escalated rows take the expert's answer to the gold text). The deployed heard view cannot evaluate a freely moving threshold: the most permissive arm escalates 50%, so the other half of each pool has no expert outcome on record, whereas the ceiling runs cover every id. Labels follow each pool's own judge (Table 1); the two judges disagree by up to 13 points on Web Questions, so they are never mixed within a row. AUC columns are recomputed from the deployed sweep's stored end-of-turn scores; they differ from Table 1's AUC column — captured in a separate scoring pass — by up to 0.03 externally and 0.04 on the internal pool (0.840 here vs. 0.879 there): the passes render and prefill the audio independently, and the probe score carries render-to-render jitter. AlpacaEval is excluded — it has no binary correctness label on either side, only a 1–5 score whose expert ceiling is degenerate (195/199 at 5).

One absolute threshold, six pools. The deployed gate adapts per pool, but only through label-free quantiles of its own score distribution. To separate what the probe's *ranking* carries from what its *scale* carries, we freeze the single absolute threshold the internal system actually deployed (τ=0.628, the balanced tier) and apply that number verbatim everywhere. **Scale does not transfer**: the same τ realizes escalation rates from 2.0% (Reasoning QA) to 48.0% (SD-QA), a 24-fold spread, so a fixed cut cannot hold an escalation budget across pools. **Ranking does**: at whatever rate it fires, fixed-τ escalation beats a random reference at the same realized rate on four of five external pools, paired-bootstrap interval excluding zero (Table 13). The exception is Reasoning QA, where τ fires on 2% of queries and there is almost nothing to measure. This is the concrete case for the label-free quantile step: it buys budget control, not discrimination.

Does the gate find failures the expert can fix? The probe is fit and read against *local failure*, but what a routing system needs is *expert benefit*: y=1 when the local model is wrong *and* the expert is right. Re-scoring the same frozen scores against this harder label costs little on the retrieval pools (Speech TriviaQA 0.796→0.782, Llama Questions 0.830→0.813) and more where the expert is itself weak (Web Questions 0.773→0.689 against a 0.864 expert ceiling; internal 0.840→0.732). The gate thus remains a usable benefit predictor out of distribution (0.63–0.81) despite never having seen an expert outcome, and the drop is largest exactly where the paper already reports a flat live curve. On Web Questions the mechanism is direct: 75.0% of local failures are expert-fixable pool-wide, but only 65.5% of the failures inside the gate's top-scoring 15% are, and that share climbs monotonically as the cut widens (0.655/0.685/0.711 at 15/30/50%) — the gate's most confident failure calls are the ones the expert also misses. Table 14 places the deployed gate against the matched-budget oracle that escalates true benefit cases first: the gate clears random at every pool and budget, capturing a fifth to three fifths of the random-to-oracle span, and the residual gap is the headroom a decode-time or expert-aware second stage would have to claim.

J Transfer latency shapes

Matched-rate precision. Precision at a fixed escalation rate is capped at base-fail/rate: on Llama Questions (base fail 0.16) the 50% cap is 0.32 and the deployed probe reaches 0.304 — 95% of the cap (recall 0.93). Quoting raw precision would misread the strongest pool as the weakest.

The median left-fold. On easy pools per-arm P50 is non-monotonic (Llama Questions 1.52→1.17→1.60/1.42 s): a verified median-of-a-mixture mechanism, not noise. The probe preferentially escalates the queries whose *local* decode is longest — on retrieval pools wrong answers

Table 13: One absolute threshold (τ=0.628, frozen on the internal pool) applied verbatim to every pool; gold-inject view. Δ is fixed-τ accuracy minus a random reference at the *same realized rate* (paired bootstrap over ids, B=10^4, seed 42). AUC columns re-score the same frozen scores against local failure and against expert benefit (y= local wrong $\wedge$ expert right, base rate in the last column). Rates and accuracies in %; AUC in decimals.

pool	n	rate	local	acc	rand	Δ 95% CI	AUC_{fail}	$\text{AUC}_{\text{benefit}}$	base
Speech TriviaQA	250	28.4	66.4	83.2	75.0	[+4.4, +12.1]	0.796	0.782	30.8
Speech Web Questions	250	36.0	56.8	72.0	67.5	[+0.8, +8.4]	0.773	0.689	32.4
Llama Questions	250	8.0	83.6	86.8	84.3	[+0.6, +4.6]	0.830	0.813	11.2
SD-QA	200	48.0	51.5	80.5	71.4	[+4.5, +13.6]	0.800	0.765	43.5
Reasoning QA (zh)	202	2.0	58.9	60.9	59.5	[−0.1, +3.5]	0.679	0.634	30.2
internal frozen test	240	35.0	38.3	65.4	57.0	[+3.8, +13.1]	0.840	0.732	53.3

Table 14: Oracle frontier at matched escalation budgets (gold-inject view): random reference / deployed quantile gate / benefit oracle, which escalates the true y=1 cases first. The gate clears random everywhere; the oracle column is the remaining headroom. Accuracies in %.

	15% budget			30% budget			50% budget		
pool	rand	gate	oracle	rand	gate	oracle	rand	gate	oracle
Speech TriviaQA	71.0	76.0	81.6	75.5	83.6	96.4	81.6	91.2	97.2
Speech Web Questions	61.2	64.0	72.0	65.7	70.0	86.8	71.6	76.4	89.2
Llama Questions	85.0	88.8	94.8	86.4	91.2	94.8	88.2	93.2	94.8
SD-QA	57.7	63.5	66.5	63.9	74.5	81.5	72.2	82.0	95.0
Reasoning QA (zh)	63.1	67.8	73.8	67.4	72.8	89.1	73.0	76.7	89.1
internal frozen test	46.3	49.2	53.3	54.3	60.8	68.3	65.0	74.2	88.3

hedge at length (decode time and answer length r=0.97), so a wrongness-selecting probe strips the slow tail as a side effect (escalated-at-15% ids: never-arm local P50 1.711 s vs. 1.198 s kept; local accuracy 0.32 vs. 0.73). Controls: the probe does not key on length (corr(score, length) = +0.05; $\approx$0 conditioned on wrongness), and on SD-QA — where wrong answers are short — the same probe selects wrongness identically and produces no fold. The mean is monotonic (1.65$\rightarrow$2.06 s); kept-local P50 falls 1.52$\rightarrow$0.43 s across tiers (Figure 15). On Reasoning QA the shape inverts usefully: the local chain-of-thought is *spoken*, so every reasoning token enters the response clock, while the expert's chain is hidden behind a flat $\sim$3.7 s round-trip — escalation truncates the latency tail (P90 13.25$\rightarrow$10.94 s) rather than fattening it.

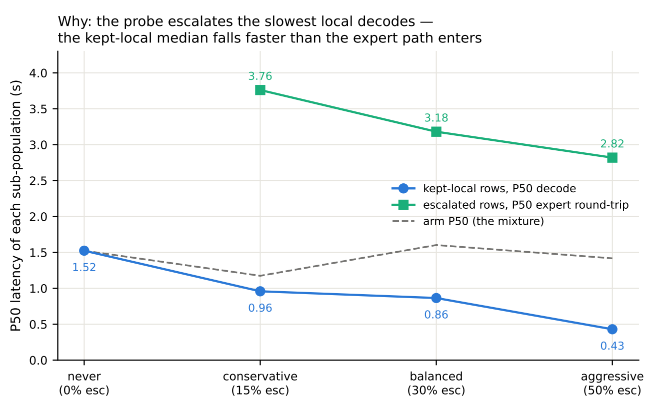


Figure 15: Mixture decomposition of Llama Questions latency per arm: kept-local P50 falls 1.52$\rightarrow$0.43 s as the probe strips the slowest local decodes while escalated rows pay a flat $\sim$3 s expert round-trip; the arm median (dashed) is their mixture.

AlpacaEval scale caveats. The official 4.8 is an offline chat-mode number — our chat-mode control reproduces it at 4.86 — and the chat-vs-live gap (4.86 vs. 3.99) is the duplex loop's spoken-reply style plus a token cap, so only live arms are comparable to each other. Selected queries score 3.90 on the never arm vs. 3.98 for skipped: no discrimination; escalated rows gain (3.90→4.71) because the expert writes better long-form answers — random picks would harvest the same gain.

K Full-pool and per-family figures

Figures 16 (dual-view) on the full frozen pool, and the complete external-family figure set (Appendix L); the noise audit behind the ±0.02–0.03 floor is Figure 17.

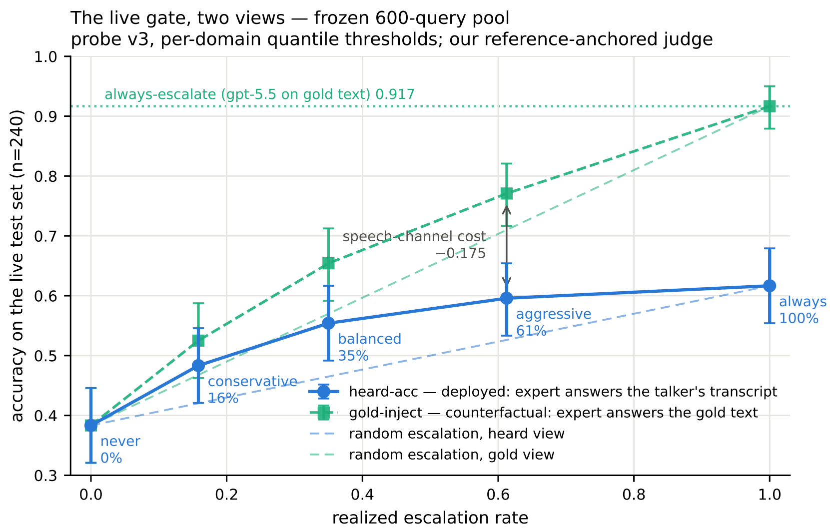


Figure 16: Dual-view live tradeoff, full pool (n=240); companion to Figure 12.

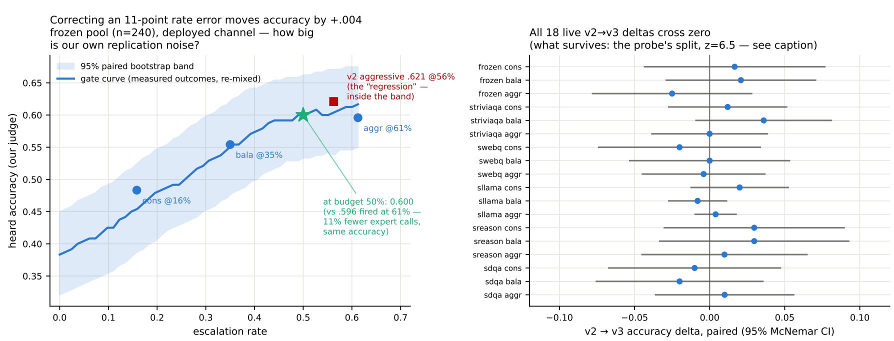


Figure 17: Replication-noise audit: same query, same audio, re-run across arms. Kept-local verdicts flip 2.3–18.8% by pool; repeated escalation 0.7–10.6%; implied paired SE 0.009–0.028.

L External-benchmark figure set

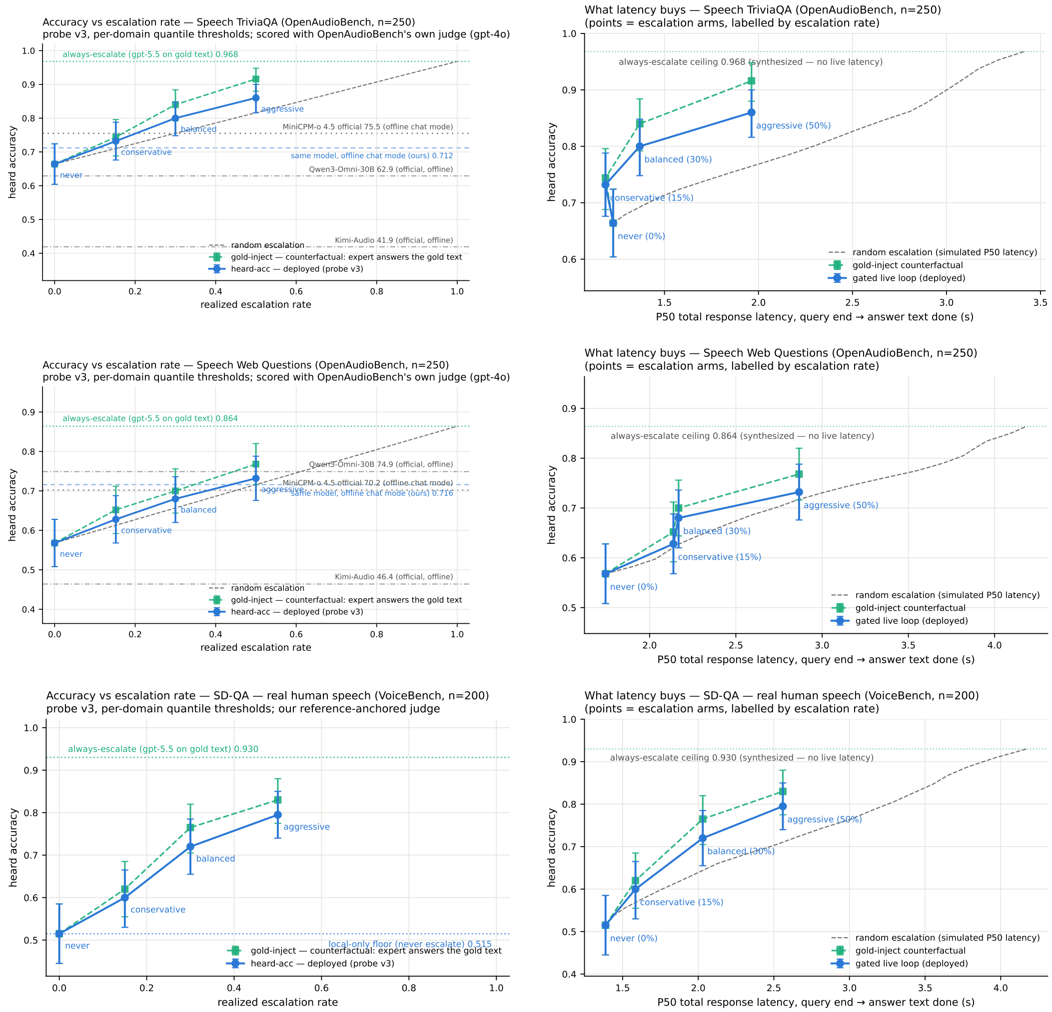


Figure 18: Speech TriviaQA, Speech Web Questions, SD-QA: dual-view (left) and accuracy–latency (right).

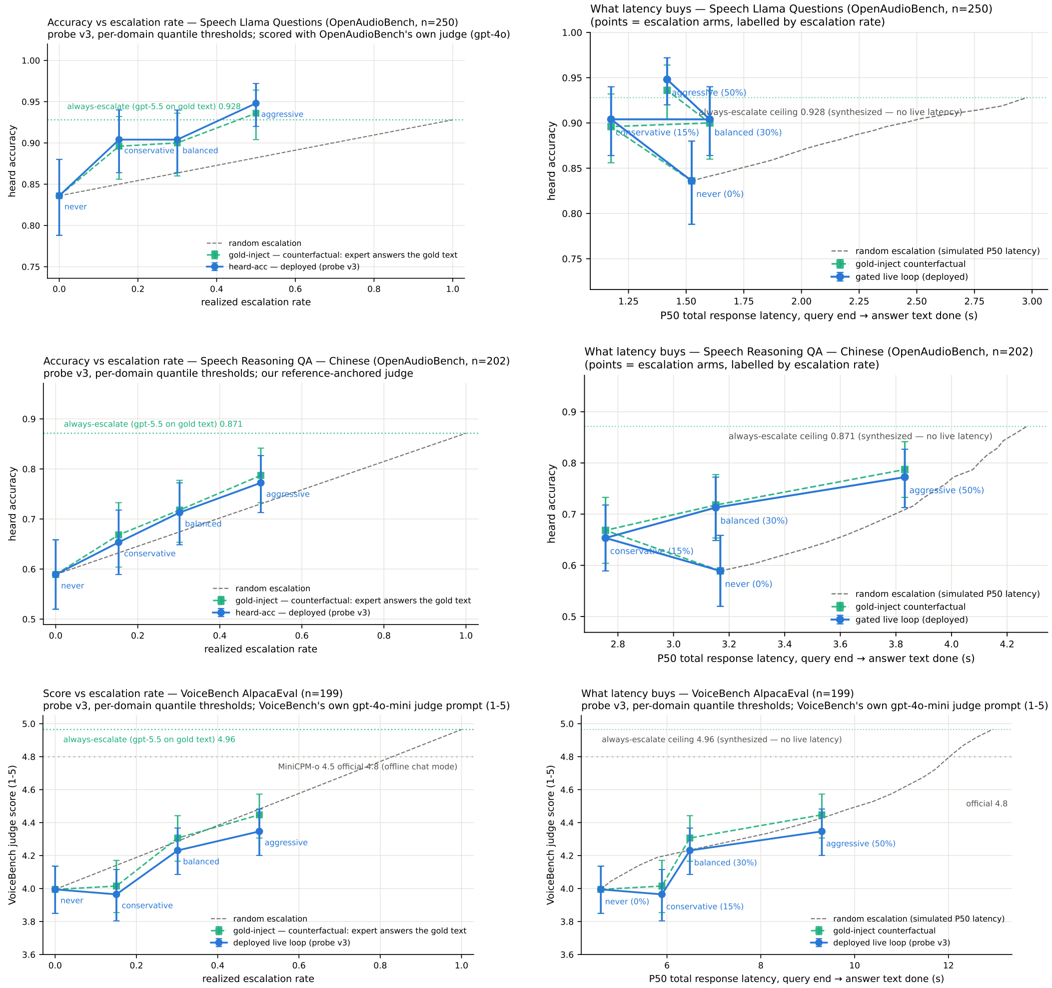


Figure 19: Llama Questions, Reasoning QA (zh), VoiceBench AlpacaEval: dual-view (left) and accuracy–latency (right).

M Judge and self-evaluation prompts

All adequacy labels use the following fixed system prompt (structured outputs with a strict {`reason`, `adequate`} schema; the judge never sees which system produced the answer):

```
You are a strict, fair grader of a small language model's answers.
You are given a QUESTION, an optional REFERENCE answer, and a
CANDIDATE answer produced by a small model.  Decide whether the
CANDIDATE is ADEQUATE, meaning ALL of:  (1) factually correct;
(2) any reasoning shown is valid; (3) it actually and completely
answers the question that was asked.  For multiple-choice questions
the candidate must land on the correct option (by letter or by
unambiguously stating the option's content).  A partially correct,
hedged, refused, off-topic, or truncated answer is NOT adequate.  If
a REFERENCE is provided, grade against it.  Be strict but do not
penalize harmless differences in format or extra correct detail.
```

The verbalized self-evaluation baselines use one fixed phrasing each; $P(\text{Yes})$ is read from the first generated token's distribution:

```
[pre] I am going to show you a question.  Do NOT answer it.  Judge
honestly whether you yourself would answer it correctly.  Question:
{q} Would you answer this question correctly?  Reply with exactly one
word:  Yes or No.
[post] Here is a question and a proposed answer.  Question:  {q}
Proposed answer:  {a} Is the proposed answer correct?  Reply with
exactly one word:  Yes or No.
```

N Reproducibility

All sampling and splits use a fixed seed (42); the calibration/test split is frozen once and the test split is touched only by the evaluation scripts. Environment pins: torch 2.8.0; transformers 4.51.0 for MiniCPM-o 4.5, with per-model pins for comparison backbones; all experiments run on single-H100 instances. Probe artifacts (weights, feature configuration, per-pool label-free thresholds), the judge prompt above, all trace files, and the per-experiment execution log — including per-run GPU and API spend, on the order of $500 total — ship in the project repository, released upon publication.

TODO (process only — delete before submission)

- [TODO: P1 — DONE 2026-08-25 (external review applied): main text now ends within 8 pages (refs start mid-page 8). Restructure: 3 contributions + hero figure; merged signal section; live 6pp $\rightarrow$ 1.2pp; router audit 3 lines; title dropped “Zero-Training”; causal claims softened (“consistent with”); no version numbers / artifact filenames in main text; everything cut lives in the appendix (A–L) or git history]

- [TODO: P2 — peer + meeting feedback (GPT reviewer pass 2026-08-26, borderline 4/7). (a) and (b) DONE 2026-08-26; (c) dropped as not computable; (d) unchanged. (a) DONE — three claims narrowed: “routing beats scaling the speech backbone” $\rightarrow$ “selective cloud escalation outperforms larger standalone backbones”; “standard readouts fail” $\rightarrow$ “late-layer readouts turn brittle after duplex fine-tuning” (abstract, intro contribution 1, hero + teaser panel-1 titles); and the EOT loop is no longer equated with native full-duplex — intro now carries an explicit scope sentence (“*duplex* names the training regime, not the interaction protocol”), live.tex states the turn-based protocol up front and is retitled “Live End-of-Turn Escalation”, and “duplex session/system” $\rightarrow$ “spoken session”/“live system” throughout. (b) DONE — new appendix section (app:fixedthr) + two tables: one absolute threshold ($\tau{=}.628$, frozen internally) applied verbatim to all pools fires at 2–48% yet still beats matched-rate random on 4/5 (ranking transfers, scale does not); expert-benefit oracle (y = local-wrong AND expert-right) with AUC re-scored per pool and a gate/random/oracle frontier at 15/30/50% budgets. Script: scripts/09_fixed_threshold_oracle.py $\rightarrow$ figures/fixed_threshold_oracle.json. Gold-inject view (heard view only observes the $\sim$50% of ids some arm escalated); labels follow each pool’s own judge (oab_ok vs adequate — they differ by 13pp on Web Questions, never mix); AlpacaEval excluded, it has no binary label on either side. (c) DONE 2026-08-26 (was DROPPED, then re-run with TTS on): bench_live gained a tts flag (init_tts + Token2wav, canned stall, gen_speak port of demo_app); balanced arm re-run on the frozen 240 ($\sim$50 min H100, expert cache hits). EOT$\rightarrow$first-audible: local p50 .552s/p99 .644s, escalated (canned stall onset) p50 .022s/p99 .029s, expert-content p50 6.4s/p95 26.0/p99 34.0. Same 84/240 fire as text-out; probe read unperturbed (p50 21ms). Landed: live.tex sentence + app:livedetail paragraph. Traces data/frozen_v3tts_traces.jsonl.balanced.shard0-3; analysis scripts/10_tts_latency.py. Fixed en route: fig:timeline-live panel (c) was captioned “first token 0.4 s” but the plotted value is answer_ms = *full* answer decode; caption + figure title corrected and the figure regenerated. (d) DONE 2026-08-27 (was SKIP). 8as closed the last tier: concurrent prefill — target audio interleaved into the SAME KV stream between generation chunks, EOT read fires while generation is active (238/240). Frozen v3 probe degrades AUC 0.869$\rightarrow$0.758 (paired Δ -0.109 [-0.158,-0.064]) — the ONE shift on the ladder that breaks frozen transfer; label-free requantiling recovers decisions only to 75%. In-regime refit of the same 12288-d read on 360 concurrent calib rows recovers 0.818 [0.761,0.871] at quarter-scale data $\Rightarrow$ signal intact, calibration must follow the regime. Landed: app:duplexval third arm, Limitations (vi), intro clause. Harness modal_duplex_concurrent.py; scripts 11/12; RESULTS 8as. Page rebalance done same pass: Table 2 policy table, dualview figure, second-family block and the serendipity paragraph moved to the appendix; main text ends p8, refs p9. Scale unified (accuracies/rates in %, AUC in decimals). Still open: official duplex serving format, full-scale (2310) in-regime recalibration, human continuity study, second expert. (d-earlier) 8aq duplex-validation sweep ran — the frozen-pool 240 replayed through the DEPLOYED voice loop (talker on, spoken answers) in clean + overlap/barge-in arms via _ws_duplex.py against the live demo; frozen v4 probe, no refit: AUC .871/.856 vs .866 headless (paired deltas inside the replication floor), decision agreement .89/.91, barged 237/239. Landed: live.tex closing paragraph, app:duplexval, intro scope sentence updated, Limitations (vi) narrowed to “concurrent-with-generation read remains open”. Data: gate-data volume duplex_sweep/{clean,overlap}.jsonl; analysis _duplex_analyze.py. This also partially recovers (c): the voice loop logs EOT$\to$first-audible per turn (first_audio_client_ms, p50 $\approx$.58 s, both arms) — usable if a reviewer asks, though only under probe-off replay, not the escalating sweep. Still skipped: human continuity study, second expert, model-level concurrent listen/speak read. Frame the paper as what it is: pre-response competence probing in duplex-trained models + turn-based live escalation, now with a deployed-loop validity check.]

- [TODO: P2.2 — meeting feedback round 2 (2026-08-26), four items, all landed same day: (1) ugly zigzag figures: fork_pareto redrawn as depth-colored scatter + Pareto frontier (figures/fork_pareto.py, new); sllama_latency_decomp cut to the clean single decomposition panel (median-vs-mean fold fact stays in prose). (2) main table: tab:transfer upgraded Kitty-style — live tiers + ceiling + AUC + official PLUS gold-inject@30% rand/gate/oracle block (numbers from tab:oracle); judge folded into row names. Views kept separate and labeled — do NOT let anyone compare live heard vs gold-inject across blocks. (3) special-token routing discussed in related work: pause/thought tokens (goyal2024pause, zelikman2024quietstar), fused think modes (yang2025qwen3, guo2025deepseekr1), MiniCPM-o's own think gate — they decide more compute not which model, need talker fine-tuning, policy baked per checkpoint; ours = frozen + pluggable + beats the built-in gate at matched budget (tab:policy). (4) simple-but-important framing: intro minimalism sentence ("competing signals read the wrong place / arrive too late / must be trained in"). Page budget paid by: fair_dualview .56$\to$.45 linewidth, caption + related + gap compressions.]
- [TODO: P2.3 — storytelling pass (advisor feedback 2026-08-26, selectively adopted): narrative arc sharpened to "looks solved $\to$ the obvious readout is exactly what duplex training corrupts $\to$ signal sits one band earlier $\to$ read it before speech commits". Abstract rewritten to that arc; intro adds the question line ("not whether one can build a router — whether the model knows it will fail early enough to do anything about it") and the find-it/read-it-in-time/show-it-matters framing; "before the first token" upgraded to "half a second before the first audible word" (grounded in the TTS-on p50 .552s + gate $\sim$30ms); conclusion closer = "the signal was never gone — only misread". NOT adopted: restructuring the three contributions, cutting appendix richness. Space paid by trimming the extended-analyses name-drop line.]
- [TODO: P0-R — reviewer P0 pass (2026-08-27), five questions; answers exist in-repo, story needs surfacing: (1) why route pre-first-token: NOT a safety necessity (first token $\to$ first audible $\approx$0.5 s exists) — it's that waiting buys no AUC (measured: entropy@4 0.696, decode-hidden mean 0.776 vs prefill read 0.828/0.887) and every ms of delay is dead air on the escalated path (expert RTT = 85–88% of escalated wall time). Edit: one sentence in gate.tex reframing pre-token as free-and-optimal, citing the measured decode-time rows. (2) final-layer mean-pool WAS run: LOPO math o4.5 $\approx$0.80 (L35) / o2.6 0.674 vs mid last-token 0.931/0.822; in-dist uniformly worse ($-$0.058 at L22). Edit: replace signal.tex "are fine" with the numbers; claim = "only uniformly robust readout", not "mid-layer necessary". Watch counter-q: audio-native final layer is intact (0.936) — answer = text side still inverts and text$\to$audio calibration only transfers mid-band. (3) label = local-wrong by design (expert-agnostic / pluggable); measured benefit-AUC cost already in app:fixedthr (0.63–0.81; internal 0.840$\to$0.732; WebQ mechanism paragraph). Edit: promote 2 sentences to main text. Optional if labels exist: benefit-trained refit on stored EOT hiddens (CPU, minutes). (4) 20/45 ms = wall-clock prefill-start$\to$L22 score, truncated forward + probe dot, CUDA-synced, H100, 50q, warmup excluded, P50 (P95 25/104); TTFT same bench. Excludes network/RPC — async path, reported separately. Edit: measurement-protocol sentence in app:gatedetail + define in gate.tex. (5) user-facing expert latency IS measured: stall onset p50 22 ms, stall 3.2–4.1 s, expert content audible p50 6.4 s/p95 26/p99 34, dead air after stall p50 $\approx$2.3–3.2 s, tail P90 $\approx$27 s. Edit: put content-audible numbers next to

the 22 ms claim in live.tex main text so 25$\times$ can't be read as answer latency; optional: exact dead-air p50/95/99 from frozen_v3tts traces (scripts/10_tts_latency.py extension, CPU). STATUS 8-27: all five LANDED (gate.tex $\times$2, signal.tex, live.tex, appendix $\times$3); dead air computed from the tts traces — p50 2.5 s / p95 22.4 / p99 30.7, 25/84 turns fully covered (scripts/10_tts_latency.py, dead-air block added). Benefit-trained refit NOT run (only re-scoring exists). $\Rightarrow$ WARNING page budget: same-engine (tectonic) check puts References at p9@78% BEFORE this pass and p10@6% after — main text already exceeded the 8-page target by $\sim$0.8 pp before these edits (user build 8-27 01:52 also shows refs on p9). Needs a $\sim$1-page trim decision before upload.]

- [TODO: P2 — 8-27 advisor pass: related.tex gained the full-duplex $\times$ special-token cell (wang2024fsm, zhang2024duplex, veluri2024syncllm, ma2024lslm, fu2024vita, yu2025salmonnomni) + explicit 2$\times$2 gap framing. Cross-check citation list against BrownCat's Feishu duplex list before camera-ready; swap in DuplexPO-style entries if the list has better exemplars.]
- [TODO: P2 — 8-27 mechanism audit (8aw) LANDED: §4.2 no longer asserts turn-control specialization — the speak-mode, modality-axis, pool-identity and drift candidates are all ruled out in D, and the surviving statement is the delivery/distributed-read one. UPDATE same day (8ax): the two direct tests RAN (control-token logit lens + listen/speak intervention, `modal_interp.py`): output layer is control-biased (log-mass rises to -11.8 while backbone falls to -38.5) but the inverting direction aligns with neither control unembeddings (cos .069 = random .068) nor the cue axis (cos .019 $\approx$ chance) — simple takeover story fails its own direct tests; appendix says so. If a reviewer pushes on mechanism, that is the honest answer.]
- [TODO: 8at DONE 2026-08-27 — the FULL loop in the concurrent regime, all 7 pools (36 arms): monotone gains everywhere, selective beats always-escalate on internal (66.2 vs 61.7) and Llama Q.; speakable 44.5$\rightarrow$71.1 %; in-regime probe AUC .857 internal / .57–.67 external (calibration-scale gap, 360 rows); latency holds (EOT 23ms, stall onset 27–34ms, local first audio .655s). Landed: app:duplexval concurrent paragraph + tab:conclive, live.tex pointer, Limitations. Page rebalance: tab:signals, tab:policy, the THINK paragraph and "What the signal detects" moved to the appendix; Conclusion is now a paragraph of the Discussion. Main text ends p8, refs p9. Known gaps for camera-ready: full-scale (2310-row) in-regime recalibration; Nemotron under interleave (SATURDAY nice-to-have, after P3 — its NeMo stack is cacheless per-frame so overlap can be constructed offline); official duplex serving format.]
- [TODO: P3 — submission day (8-28): delete this section; final anonymization grep (names, rhe9527, modal URLs, gmail); decide public-repo $\rightarrow$ private; OpenReview form + upload (deadline 8-29 AoE, moderation hard limit Fri 8-28)]